IT Security Assessment Frameworks by Country

A Multi-Framework Reference for SA&A Tool Expansion

> **Scope:** This document covers IT-focused security assessment and authorization frameworks used by governments and regulated sectors across Canada, the United States, Europe, and Australia. Frameworks are limited to those with a structured IT security control assessment lifecycle compatible with SA&A-style tooling (control catalogues, risk categorization, evidence collection, authorization decisions, and continuous monitoring).

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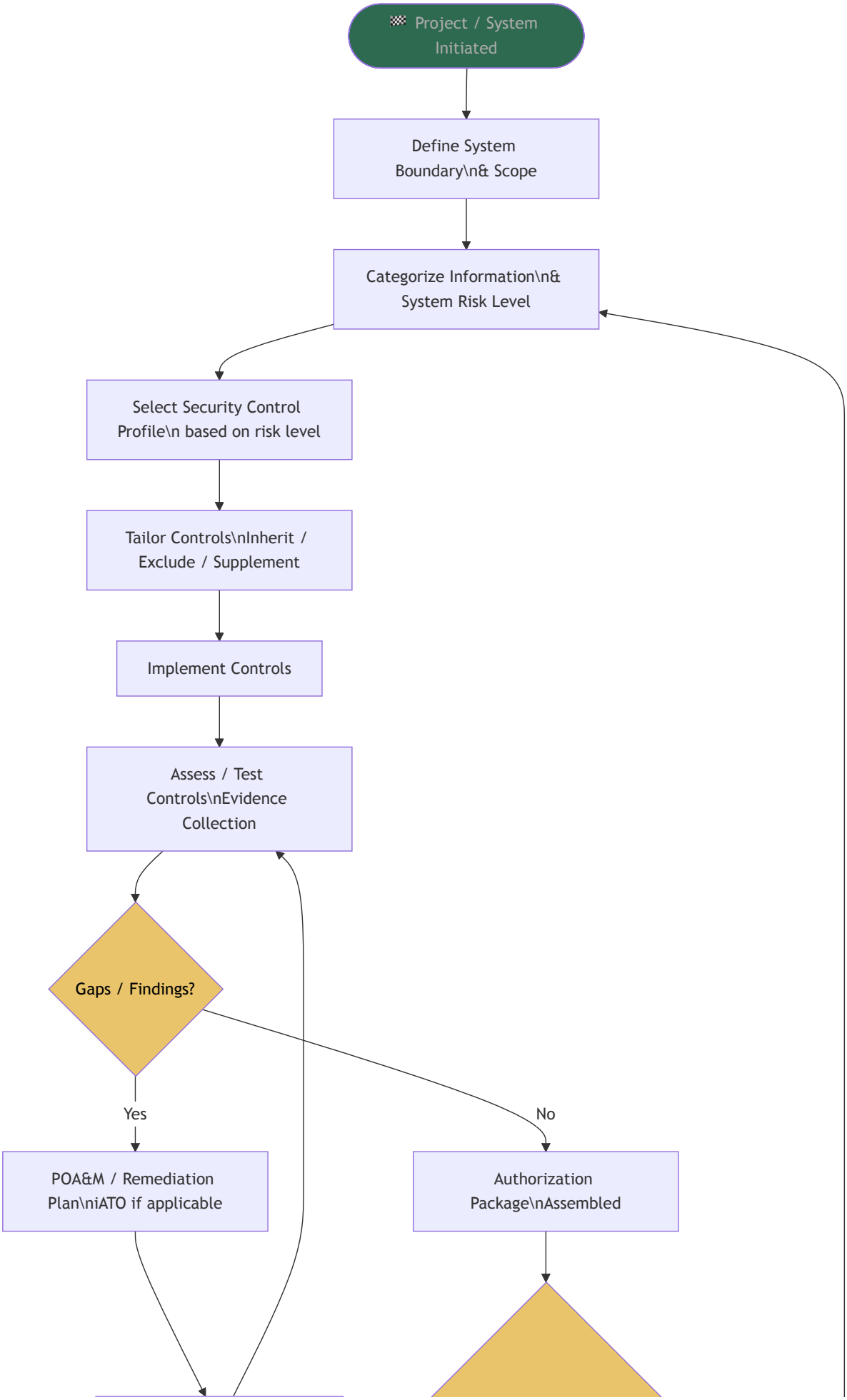
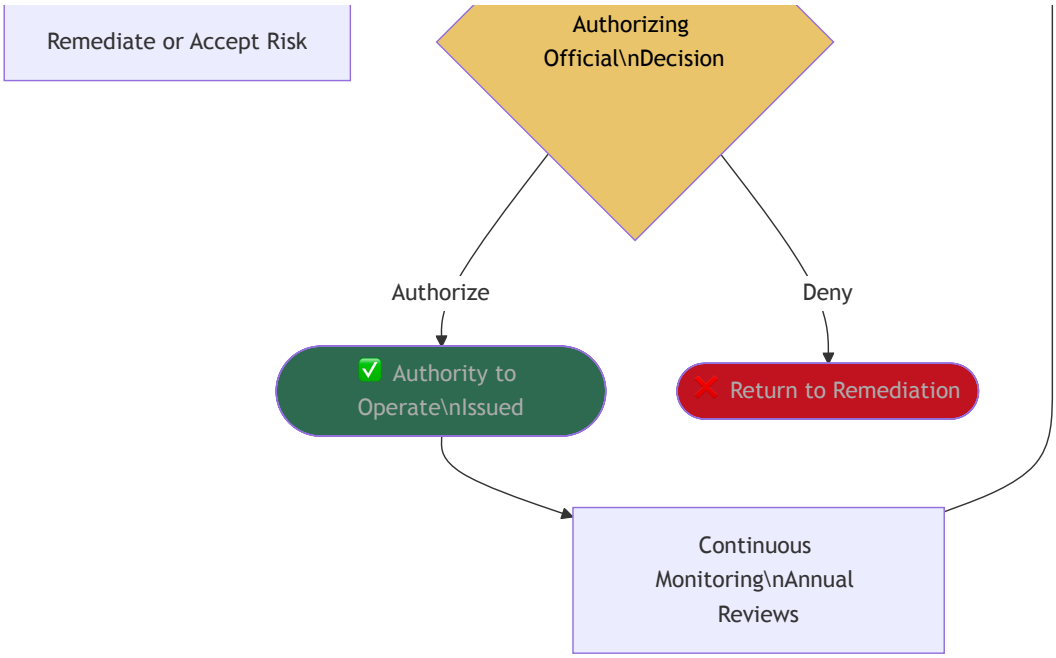
1. Common SA&A Lifecycle (Universal Model)

Despite differences in terminology and control catalogues, all IT security assessment frameworks share the same fundamental lifecycle. This is the core engine your tool already implements for ITSG-33.


Project / System Initiated
Define System Boundary\n& Scope
Categorize Information\n& System Risk Level
Select Security Control Profile\n based on risk level
Tailor Controls\nInherit / Exclude / Supplement
Implement Controls
Assess / Test Controls\nEvidence Collection
Gaps / Findings?
Yes
No
POA&M / Remediation Plan\niATO if applicable
Authorization Package\nAssembled



2. Canada

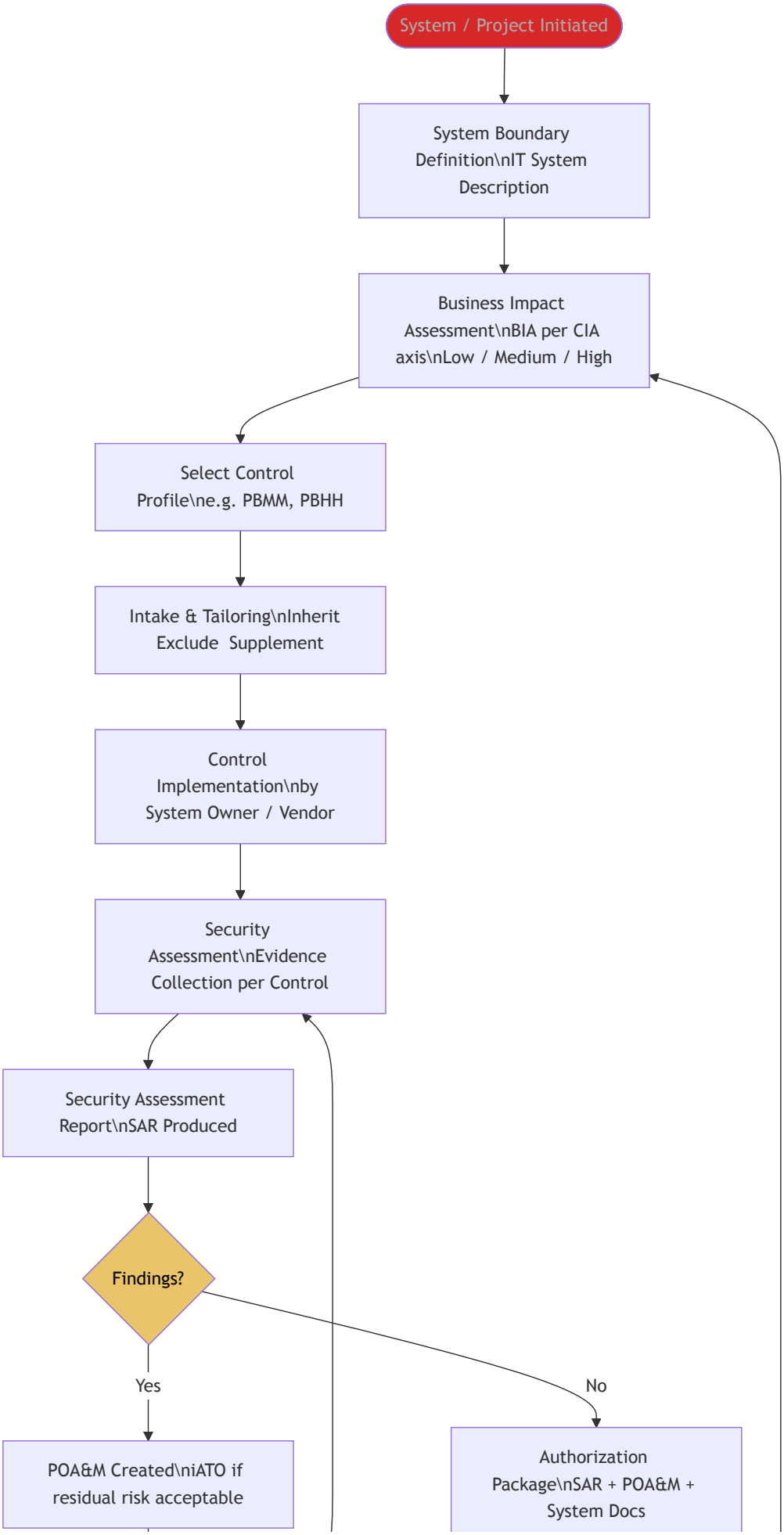
2.1 ITSG-33 — IT Security Risk Management

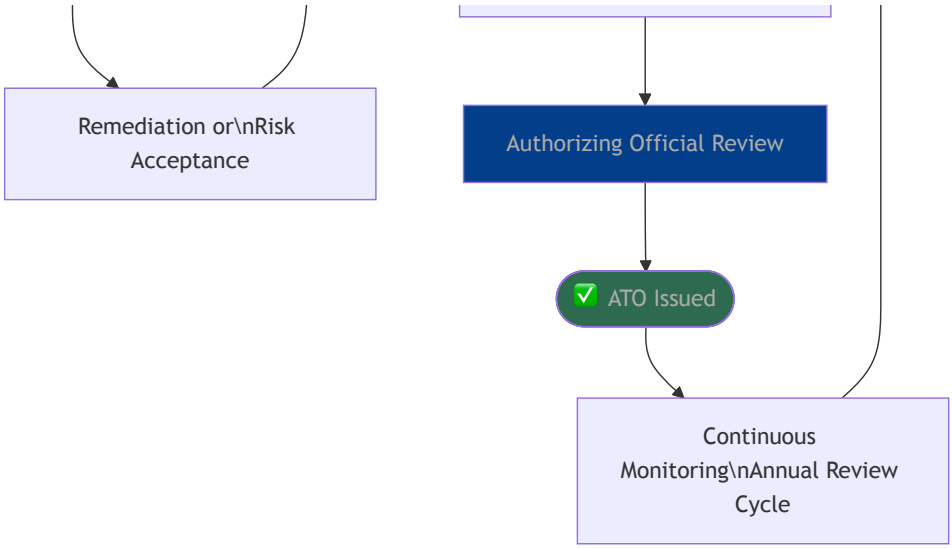
Issuing Body: Canadian Centre for Cyber Security (CCCS) / Treasury Board of Canada Secretariat (TBS) **Applies To:** All Government of Canada (GoC) federal departments and agencies **Legislative Basis:** *Policy on Government Security* (TBS), *Directive on Security Management* **Level:** Federal

ITSG-33 is the primary GoC framework for IT security risk management. It defines a full SA&A lifecycle including system categorization (based on Confidentiality/Integrity/Availability business impact), control profile selection, assessment, and Authority to Operate (ATO) issuance. Controls are organized into 18 families aligned to NIST SP 800-53.

Key Concepts:

- Business Impact Levels: Low / Medium / High per CIA triad axis
- Control Profiles: PBMM (Protected B / Medium / Medium) is the most common cloud baseline
- Roles: System Owner, IT Security Coordinator, Authorizing Official (AO), Assessor
- Outputs: Security Assessment Report (SAR), ATO Letter, POA&M


System / Project Initiated
System Boundary Definition\nIT System Description
Business Impact Assessment\nBIA per CIA axis\nLow / Medium / High
Select Control Profile\ne.g. PBMM, PBHH
Intake & Tailoring\nInherit Exclude Supplement
Control Implementation\nby System Owner / Vendor
Security Assessment\nEvidence Collection per Control
Security Assessment Report\nSAR Produced
Findings?
Yes
No
POA&M Created\niATO if residual risk acceptable
Authorization Package\nSAR + POA&M + System Docs



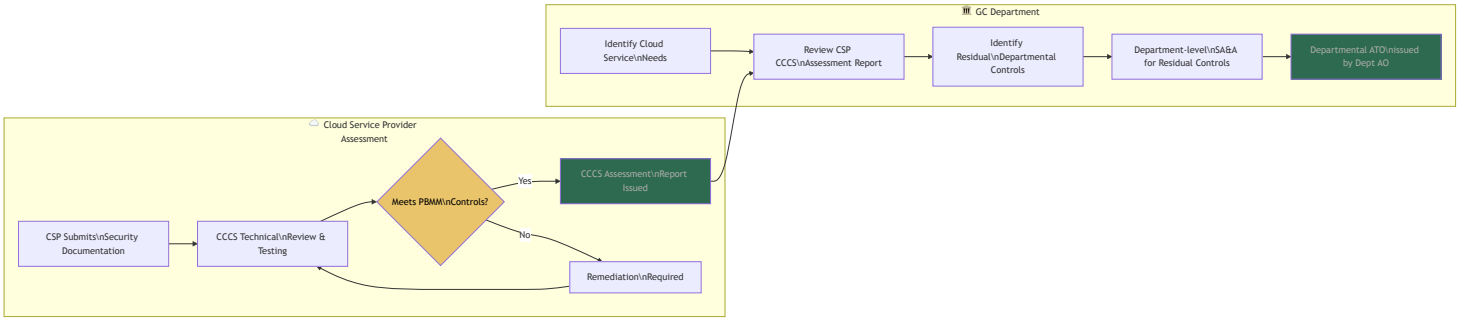
2.2 CCCS Medium Profile (PBMM)

Issuing Body: Canadian Centre for Cyber Security **Applies To:** Cloud service providers handling GoC Protected B data **Legislative Basis:** TBS *Direction on the Secure Use of Commercial Cloud Services (SPIN 2017-01)* **Level:** Federal (Cloud)

The CCCS Medium Profile superseded the original GC Cloud Control Profile. It is a tailored subset of ITSG-33 controls specifically for cloud environments targeting Protected B / Medium Integrity / Medium Availability. Cloud service providers (CSPs) undergo assessment by CCCS before being listed as approved for GC use.

Key Concepts:

- Based on ITSG-33 control families but scoped to cloud delivery models (IaaS/PaaS/SaaS)
- CSPs assessed independently; departments inherit cloud baseline and assess residual controls
- Shared Responsibility Model is central
- Aligns with FedRAMP Moderate (US) for mutual recognition purposes



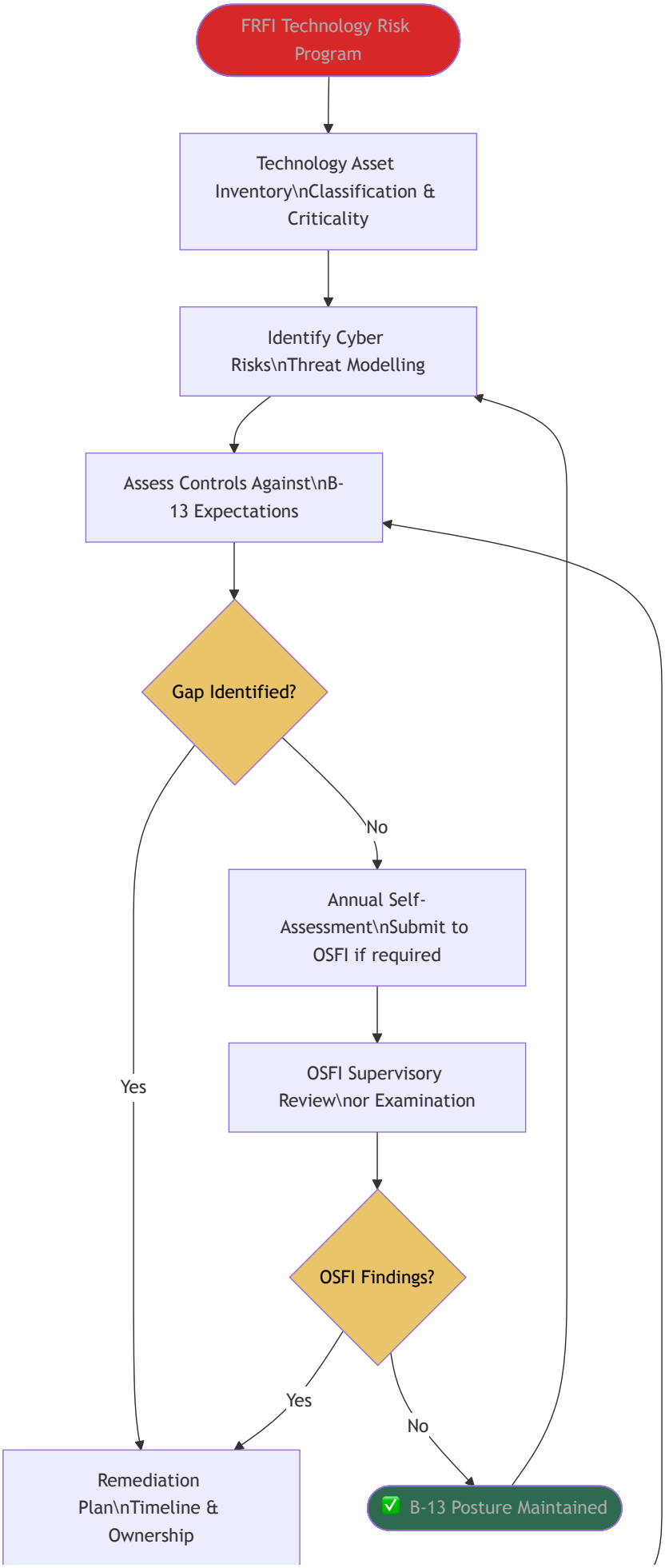
2.3 OSFI Guideline B-13 — Technology and Cyber Risk Management

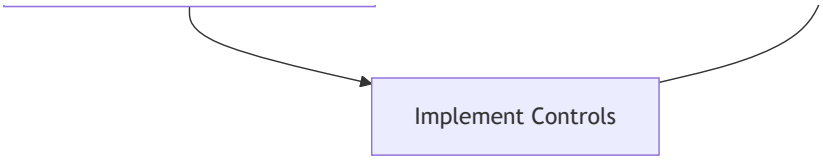
Issuing Body: Office of the Superintendent of Financial Institutions (OSFI) **Applies To:** All Federally Regulated Financial Institutions (FRFIs) — banks, insurers, trust companies **Legislative Basis:** *Office of the Superintendent of Financial Institutions Act* , *Bank Act* , *Insurance Companies Act* **Level:** Federal / Financial Sector

B-13 is the primary IT and cyber risk management standard for Canadian banks and insurers. Unlike a traditional SA&A framework, it uses an outcomes-based approach organized around Technology Operations, Cyber Security, and Technology Risk Governance. FRFIs must assess their posture against B-13 expectations and demonstrate continuous improvement.

Key Domains:

1. Governance & Accountability
2. Technology Asset Management
3. Technology Operations
4. Cyber Security (Identify / Protect / Detect / Respond / Recover)
5. Technology Risk Management


FRFI Technology Risk Program
Technology Asset Inventory\nClassification & Criticality
Identify Cyber Risks\nThreat Modelling
Assess Controls Against\nB-13 Expectations
Gap Identified?
No
Yes
Annual Self-Assessment\nSubmit to OSFI if required
OSFI Supervisory Review\nor Examination
OSFI Findings?
Yes
No
Remediation Plan\nTimeline & Ownership
B-13 Posture Maintained



3. United States

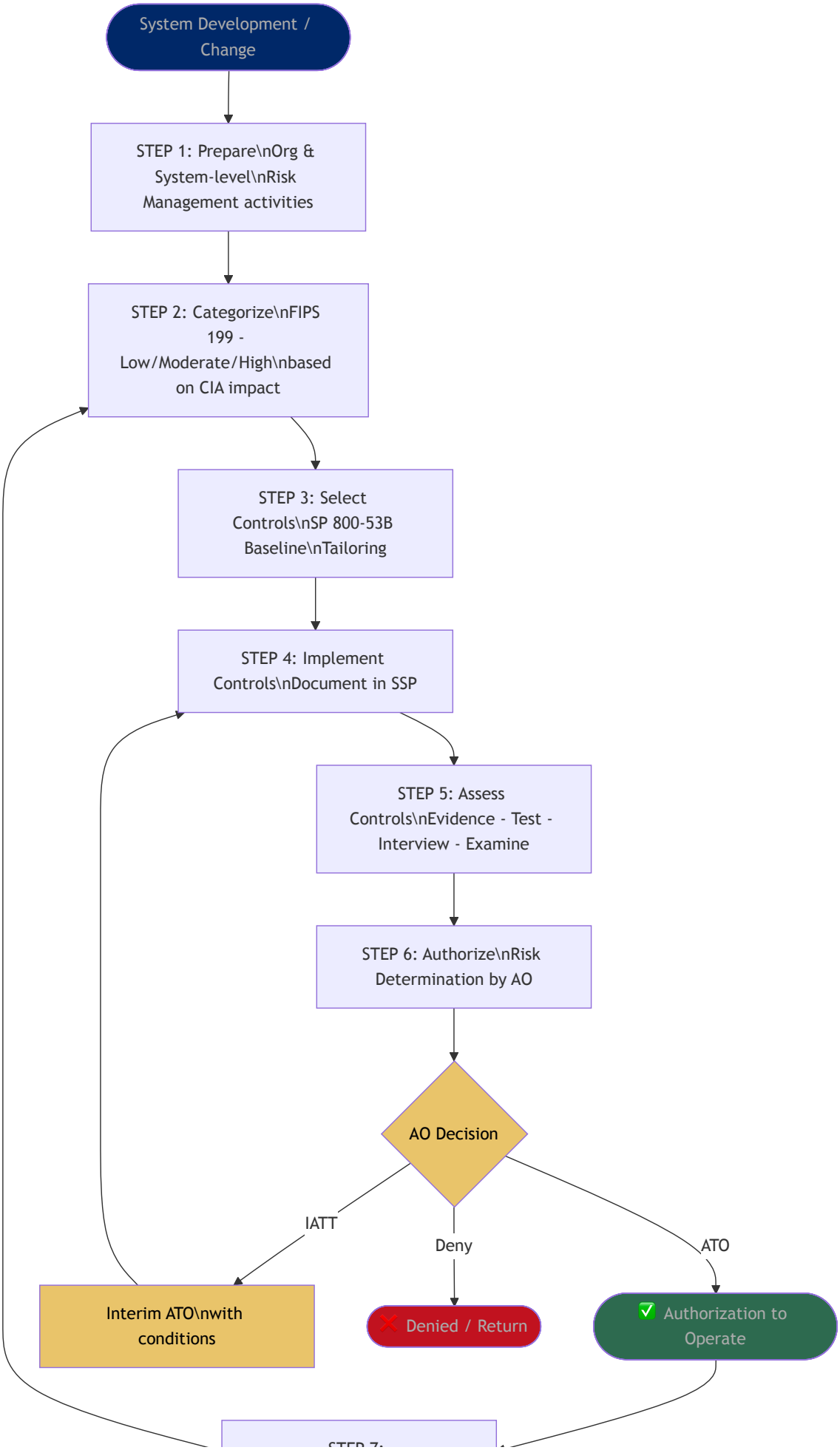
3.1 NIST RMF + SP 800-53 Rev. 5

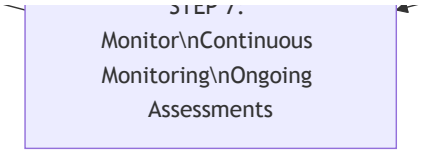
Issuing Body: National Institute of Standards and Technology (NIST) **Applies To:** All US Federal agencies and their information systems **Legislative Basis:** *Federal Information Security Modernization Act (FISMA) 2014* **Level:** Federal

The NIST Risk Management Framework (SP 800-37 Rev. 2) is the US federal equivalent of ITSG-33. It defines a 7-step lifecycle for categorizing systems, selecting controls from SP 800-53, assessing them, and issuing an Authority to Operate (ATO). SP 800-53 Rev. 5 contains 20 control families with over 1,000 controls and control enhancements. The process maps almost directly to your existing ITSG-33 tool architecture.

Key Concepts:

- System categorization via FIPS 199 (Low / Moderate / High)
- Control baselines: SP 800-53B (Low / Moderate / High baselines)
- Roles: System Owner, ISSO, ISSM, Authorizing Official (AO), Security Control Assessor (SCA)
- Privacy controls integrated into SP 800-53 Rev. 5
- OSCAL support for machine-readable control assessment data


System Development / Change
STEP 1: Prepare\nOrg & System-level\nRisk Management activities
STEP 2: Categorize\nFIPS 199 - Low/Moderate/High\nbased on CIA impact
STEP 3: Select Controls\nSP 800-53B Baseline\nTailoring
STEP 4: Implement Controls\nDocument in SSP
STEP 5: Assess Controls\nEvidence - Test - Interview - Examine
STEP 6: Authorize\nRisk Determination by AO
AO Decision
IATT
Deny
ATO
Interim ATO\nwith conditions
Denied / Return
Authorization to Operate



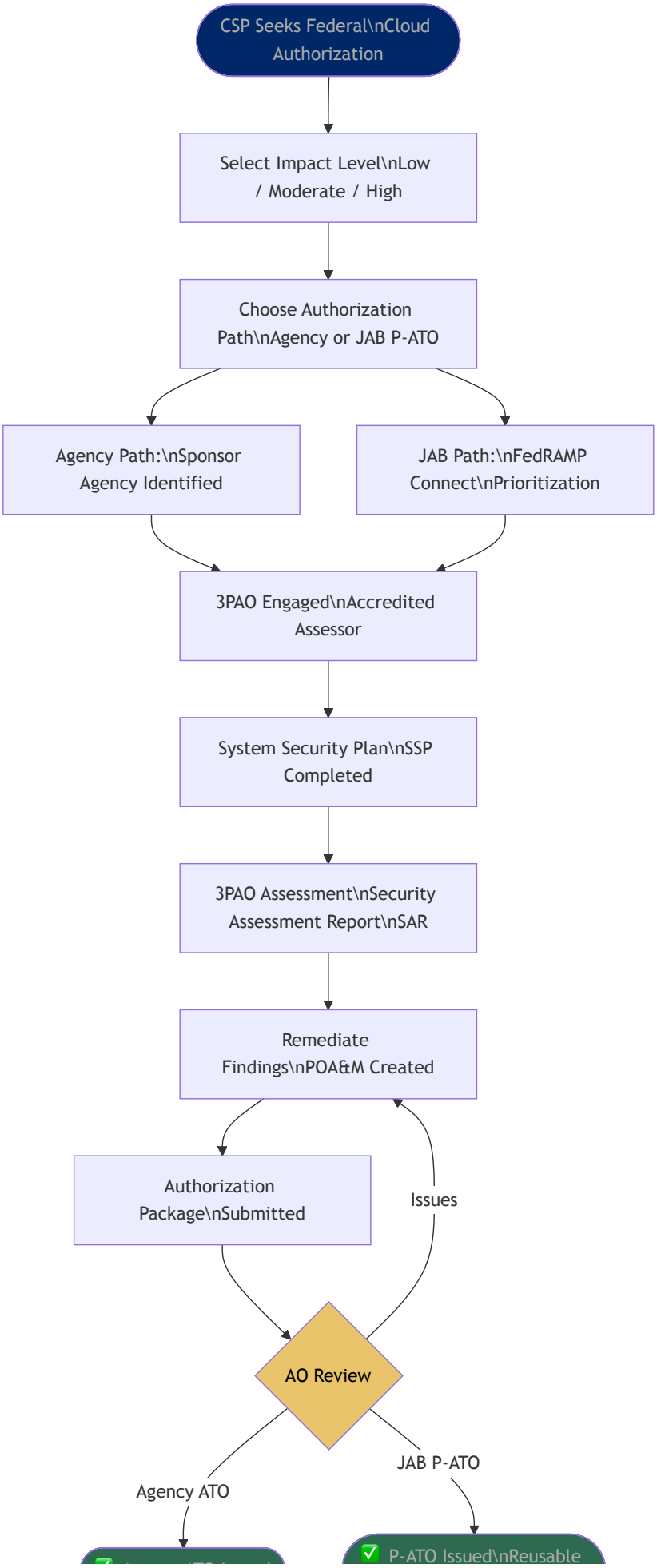
3.2 FedRAMP

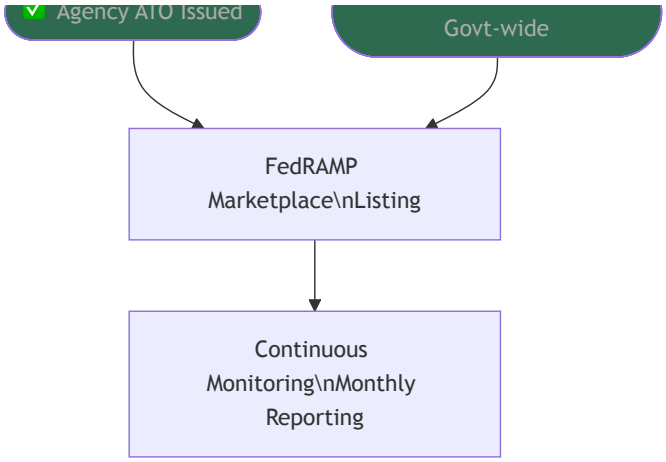
Issuing Body: General Services Administration (GSA) / FedRAMP PMO **Applies To:** Cloud Service Providers (CSPs) offering services to US Federal agencies **Legislative Basis:** *FedRAMP Authorization Act (2022)* , *OMB Memo M-11-33* **Level:** Federal (Cloud)

FedRAMP is built on NIST SP 800-53 controls and follows a “do once, use many times” philosophy. CSPs achieve authorization at one of three impact levels (Low, Moderate, High) and federal agencies can reuse the authorization package rather than conducting independent assessments. Assessments are performed by accredited Third Party Assessment Organizations (3PAOs).

Authorization Paths:

- **Agency Authorization** — a single agency sponsors and authorizes a CSP
- **JAB Provisional ATO (P-ATO)** — Joint Authorization Board (GSA/DoD/DHS) issues a provisional authorization reusable by any agency


CSP Seeks Federal\nCloud Authorization
Select Impact Level\nLow / Moderate / High
Choose Authorization Path\nAgency or JAB P-ATO
Agency Path:\nSponsor Agency Identified
JAB Path:\nFedRAMP Connect\nPrioritization
3PAO Engaged\nAccredited Assessor
System Security Plan\nSSP Completed
3PAO Assessment\nSecurity Assessment Report\nSAR
Remediate Findings\nPOA&M Created
Authorization Package\nSubmitted
Issues
AO Review
Agency ATO
JAB P-ATO
P-ATO Issued\nReusable



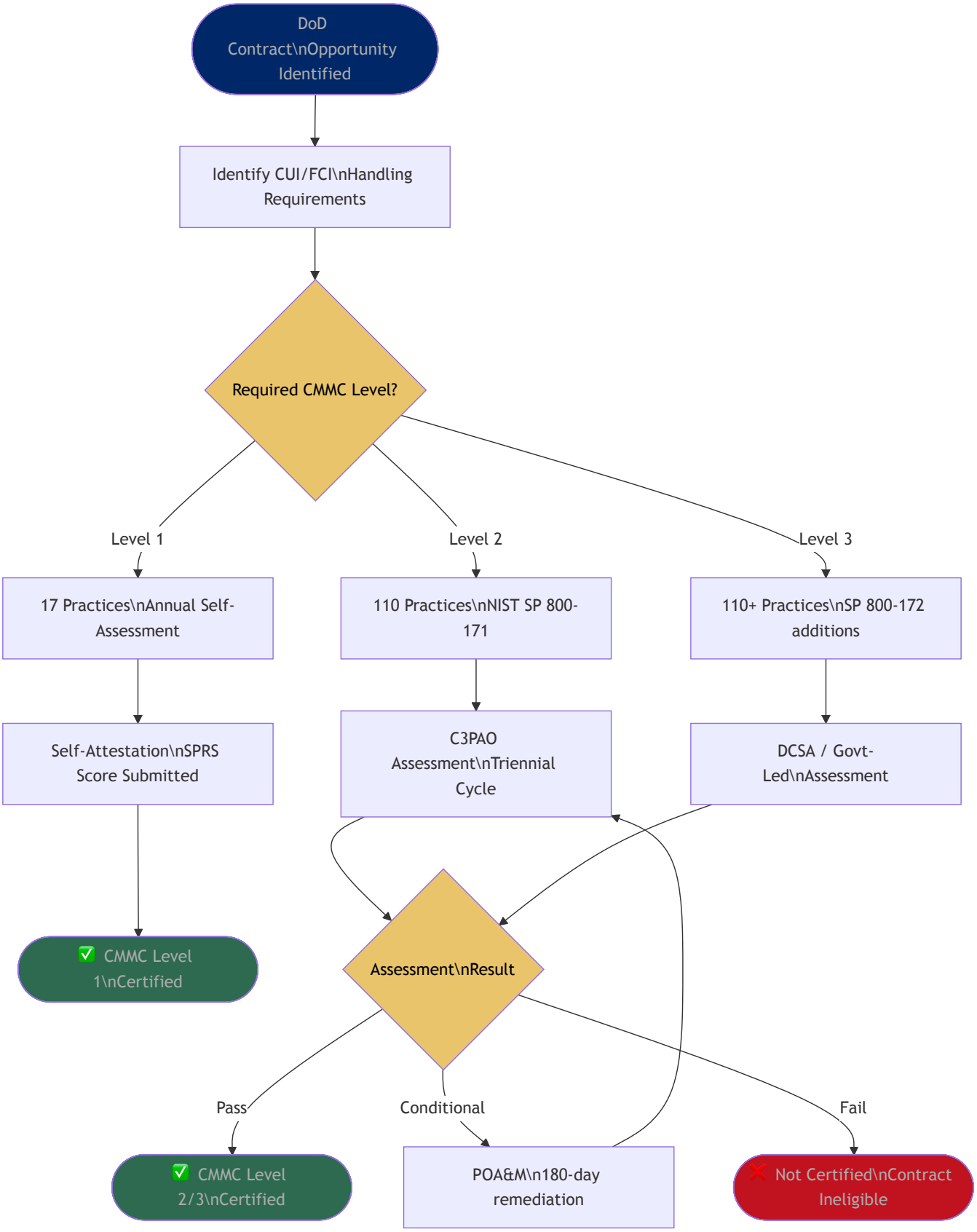
3.3 CMMC 2.0 — Cybersecurity Maturity Model Certification

Issuing Body: US Department of Defense (DoD) **Applies To:** Defense Industrial Base (DIB) contractors and subcontractors handling CUI/FCI **Legislative Basis:** *DFARS 252.204-7021* , Code of Federal Regulations 32 CFR Part 170 **Level:** Federal / Defense Sector

CMMC 2.0 certifies that organizations handling Controlled Unclassified Information (CUI) or Federal Contract Information (FCI) meet required cybersecurity practices. Based on NIST SP 800-171 (derived from SP 800-53). Three maturity levels. Third-party assessments required for Level 2 and above when handling sensitive CUI.

Maturity Levels:

- **Level 1 (Foundational):** 17 practices, annual self-assessment
- **Level 2 (Advanced):** 110 practices from NIST SP 800-171, triennial 3rd party assessment
- **Level 3 (Expert):** 110+ practices including SP 800-172, government-led assessment

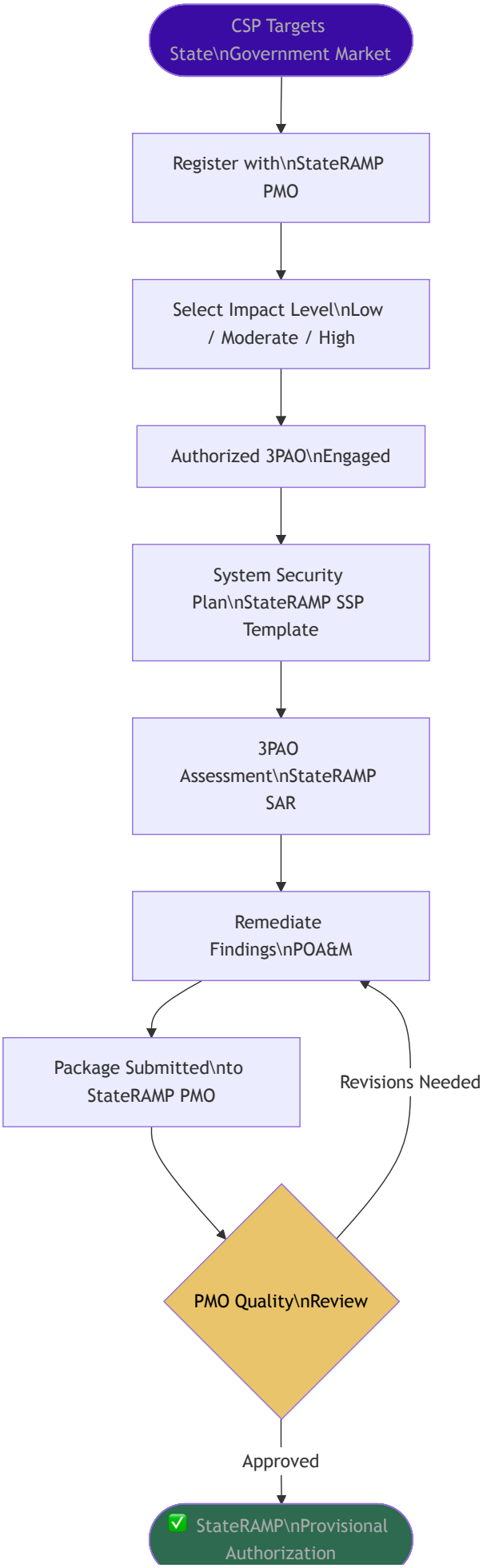


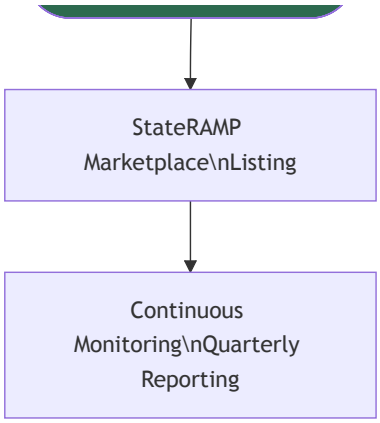
3.4 StateRAMP

Issuing Body: StateRAMP (nonprofit standards body) **Applies To:** Cloud Service Providers working with US state and local governments **Legislative Basis:** Adopted by Texas (HB 3834), Illinois, Colorado and others via state law or procurement policy **Level:** State Government

(Cloud)

StateRAMP mirrors FedRAMP's model for the state/local government market. CSPs achieve authorization at Low, Moderate, or High impact levels. A Provisional Authorization is reusable across member states. Growing rapidly as states modernize procurement requirements.

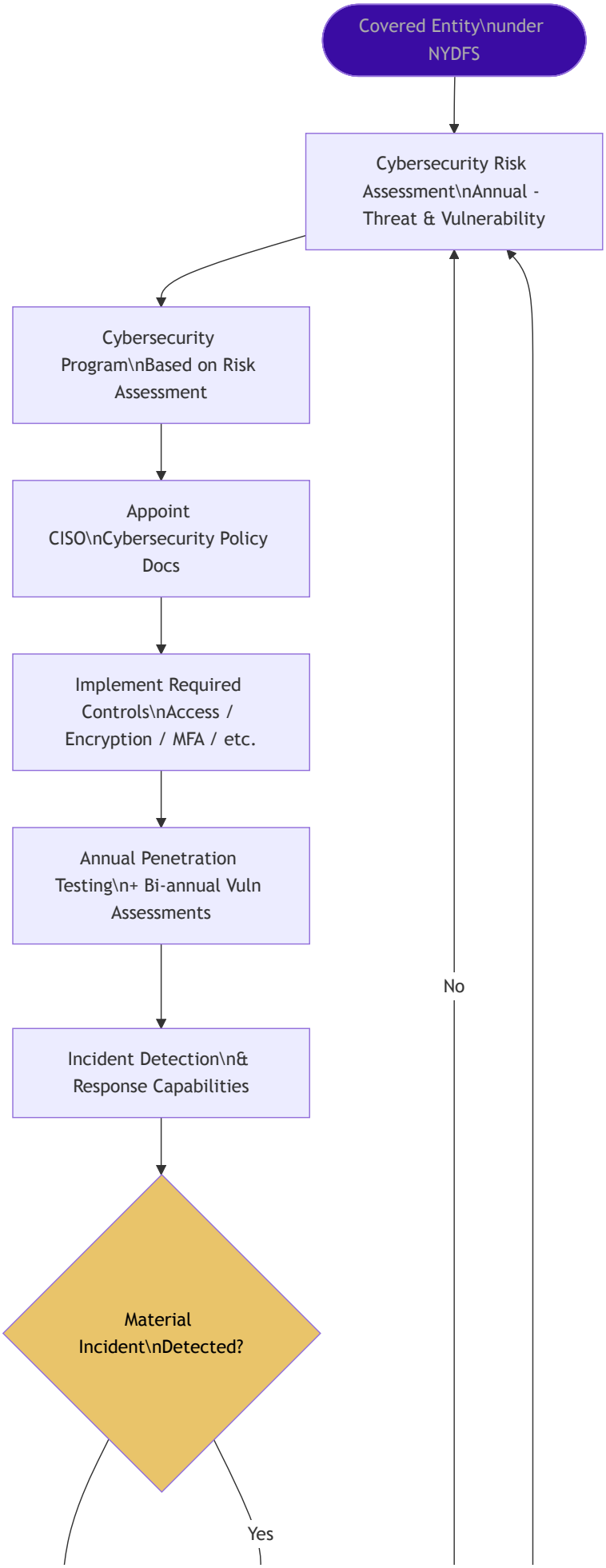

CSP Targets State\nGovernment Market
Register with\nStateRAMP PMO
Select Impact Level\nLow / Moderate / High
Authorized 3PAO\nEngaged
System Security Plan\nStateRAMP SSP Template
3PAO Assessment\nStateRAMP SAR
Remediate Findings\nPOA&M
Package Submitted\nto StateRAMP PMO
Revisions Needed
PMO Quality\nReview
Approved
StateRAMP\nProvisional Authorization

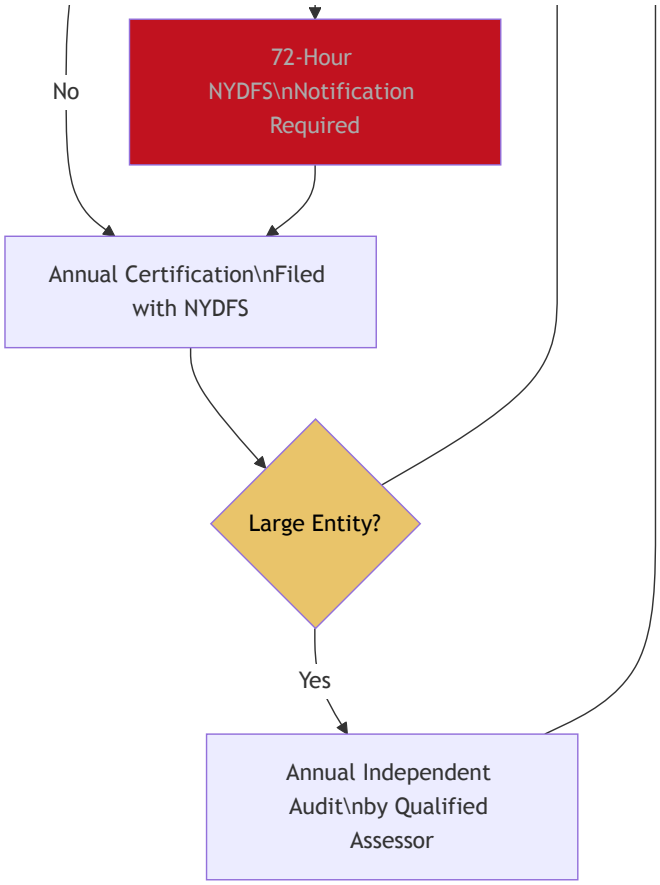


3.5 NYDFS 23 NYCRR Part 500

Issuing Body: New York Department of Financial Services (NYDFS) **Applies To:** All NY-chartered banks, insurers, mortgage companies, and licensed financial services entities **Legislative Basis:** *New York Banking Law* , *Insurance Law* ; regulation effective 2017, amended 2023 **Level:** State / Financial Sector

One of the most comprehensive state-level financial cybersecurity regulations in the US. Requires a formal cybersecurity program, risk assessments, a CISO, annual penetration testing, and specific incident notification timelines. The 2023 amendments added requirements for large covered entities including annual independent audits.


Covered Entity\nunder NYDFS
Cybersecurity Risk Assessment\nAnnual - Threat & Vulnerability
Cybersecurity Program\nBased on Risk Assessment
Appoint CISO\nCybersecurity Policy Docs
Implement Required Controls\nAccess / Encryption / MFA / etc.
Annual Penetration Testing\n+ Bi-annual Vuln Assessments
Incident Detection\n& Response Capabilities
Material Incident\nDetected?
No
Yes



4. Europe

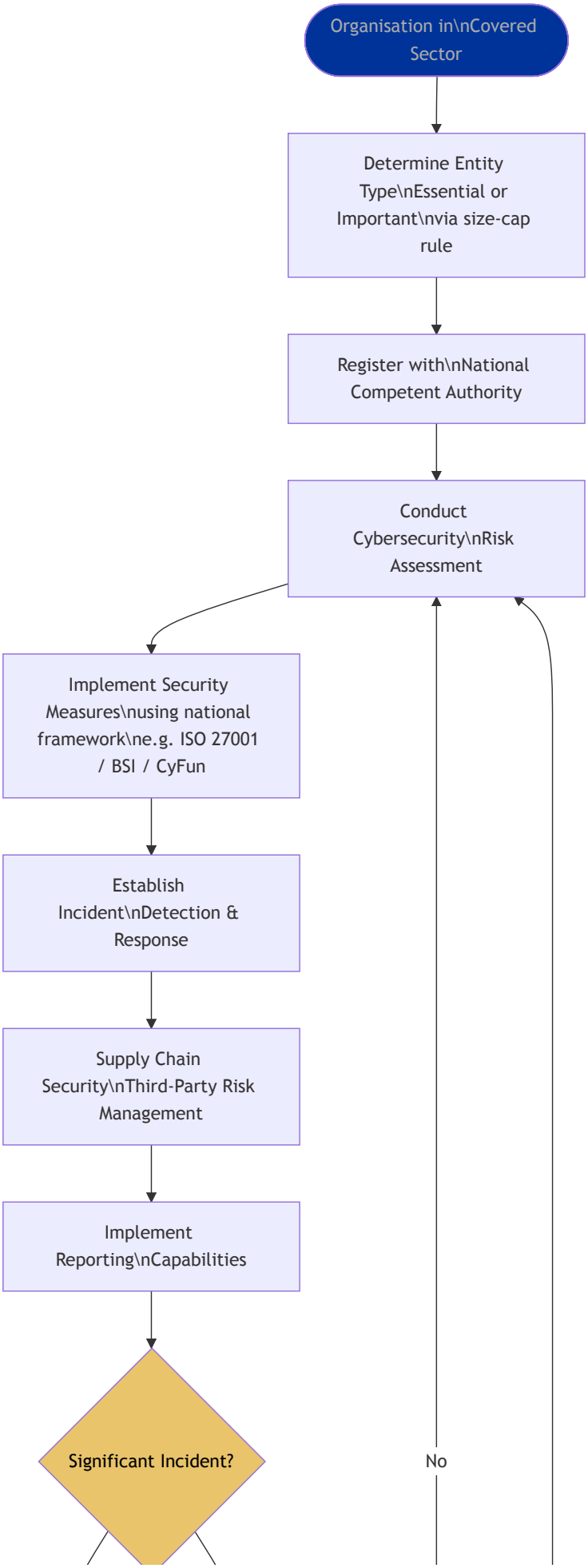
4.1 NIS2 Directive (EU 2022/2555)

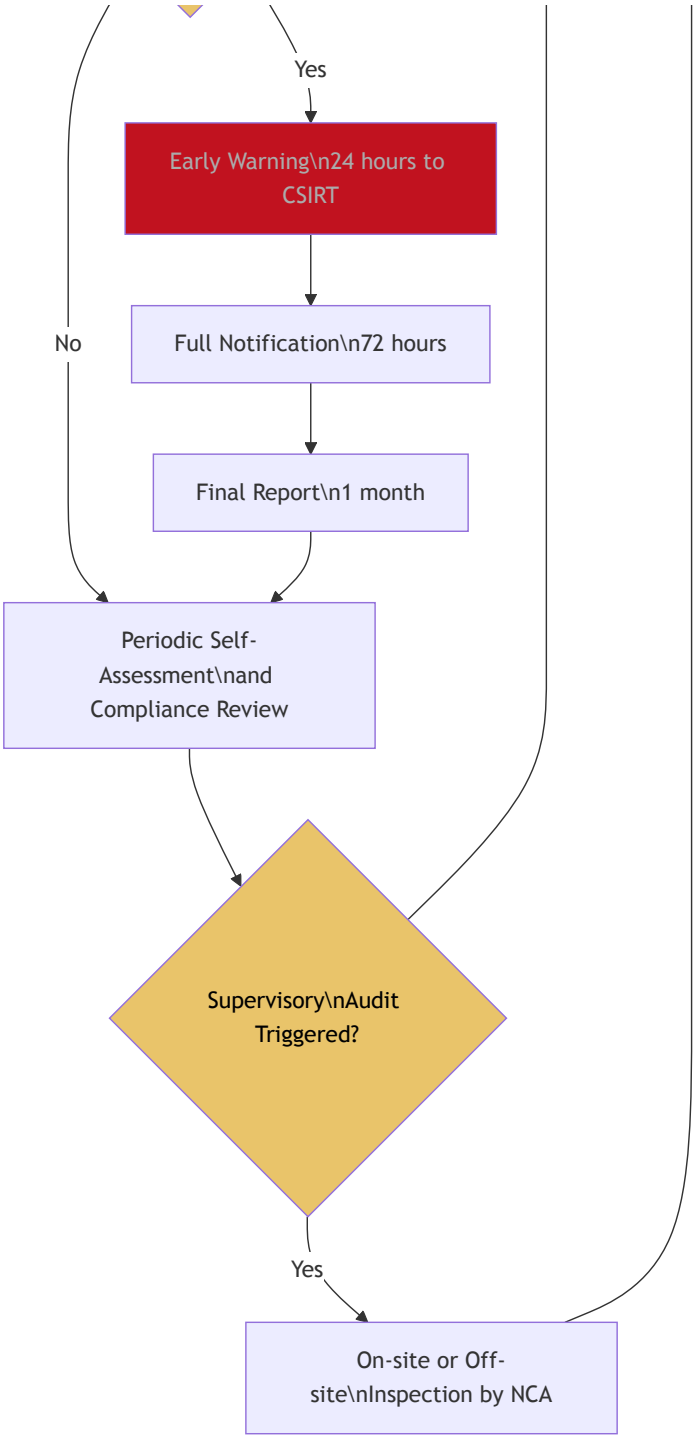
Issuing Body: European Commission / Transposed by each EU Member State **Applies To:** Essential and Important entities across 18 critical sectors including public administration, digital infrastructure, finance, health, energy, transport **Legislative Basis:** EU Directive 2022/2555, transposed into national law by October 2024 **Level:** EU-wide / National (varies by member state)

NIS2 is the EU's primary cybersecurity risk management directive. It does not prescribe a specific control catalogue but mandates that organizations implement proportionate technical and organizational security measures. Member states reference ISO 27001, NIST CSF, or national frameworks (e.g., BSI IT-Grundschutz in Germany, CyFun in Belgium) as the means of compliance. Covered entities must register with national authorities, conduct risk assessments, and report significant incidents.

Entity Classification:

- **Essential Entities:** Energy, transport, banking, financial market infrastructure, health, drinking water, digital infrastructure, public administration, space
- **Important Entities:** Postal, waste management, manufacturing, food, chemicals, research, digital providers


Organisation in\nCovered Sector
Determine Entity Type\nEssential or Important\nvia size-cap rule
Register with\nNational Competent Authority
Conduct Cybersecurity\nRisk Assessment
Implement Security Measures\nusing national framework\ne.g. ISO 27001 / BSI / CyFun
Establish Incident\nDetection & Response
Supply Chain Security\nThird-Party Risk Management
Implement Reporting\nCapabilities
Significant Incident?
No



4.2 DORA — Digital Operational Resilience Act (EU 2022/2554)

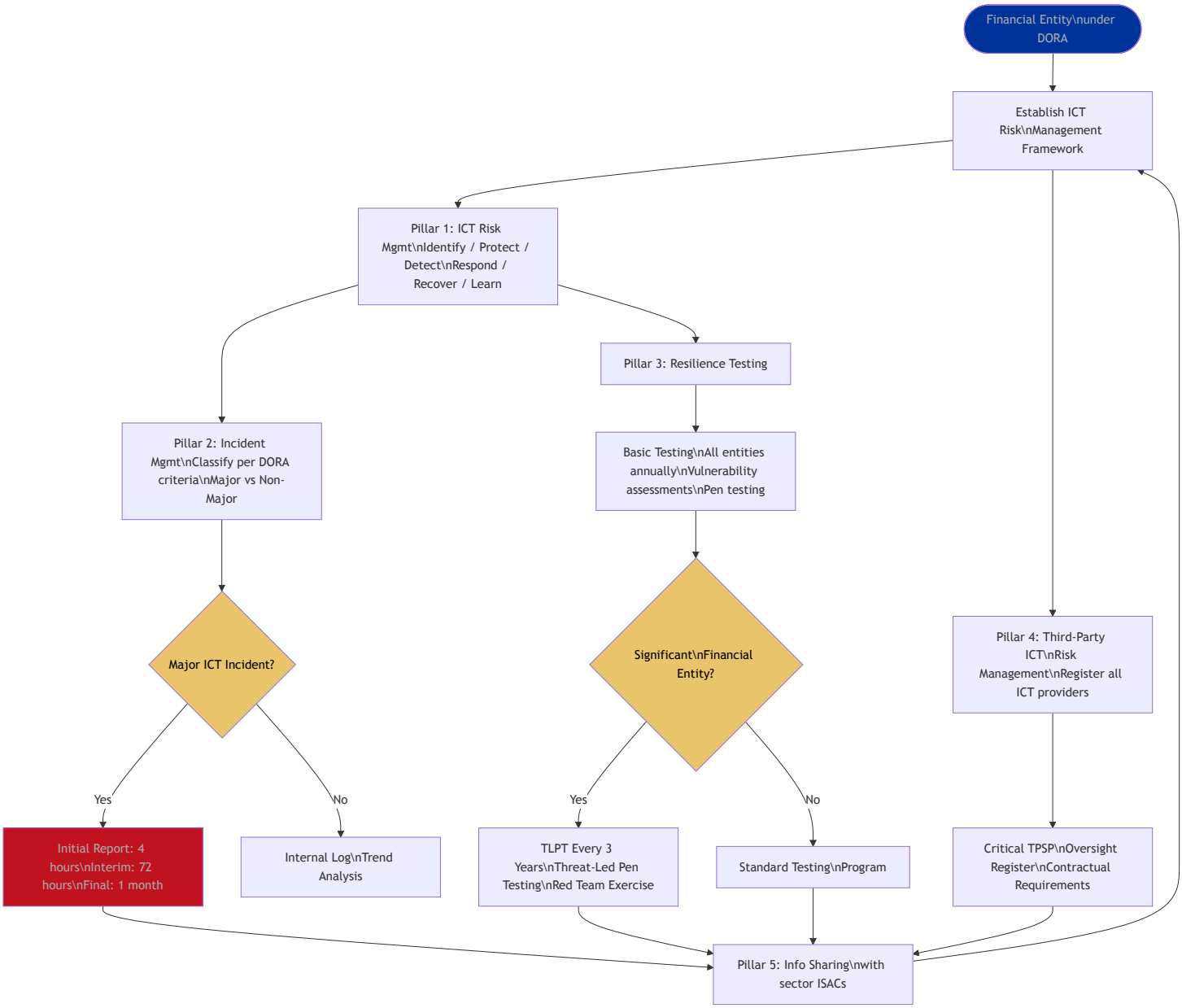
Issuing Body: European Commission (enforced by ESAs: EBA, EIOPA, ESMA) **Applies To:** Banks, payment institutions, investment firms, insurance companies, crypto-asset service providers, and their critical ICT third-party providers operating in the EU **Legislative Basis:** EU Regulation 2022/2554 (directly applicable, no transposition needed), enforceable from January 17, 2025 **Level:** EU-wide / Financial Sector

DORA is the most detailed EU IT security regulation for financial services. Unlike NIS2 (a directive), DORA is a regulation — directly enforceable in all member states with no national variation. It mandates a full ICT Risk Management Framework with five pillars. The Threat-Led Penetration Testing (TLPT) requirement is particularly notable — similar in spirit to a formal security assessment and authorization process.

Five Pillars:

1. ICT Risk Management Framework

2. ICT Incident Classification, Reporting & Management
3. Digital Operational Resilience Testing (including TLPT)
4. ICT Third-Party Risk Management
5. Information Sharing

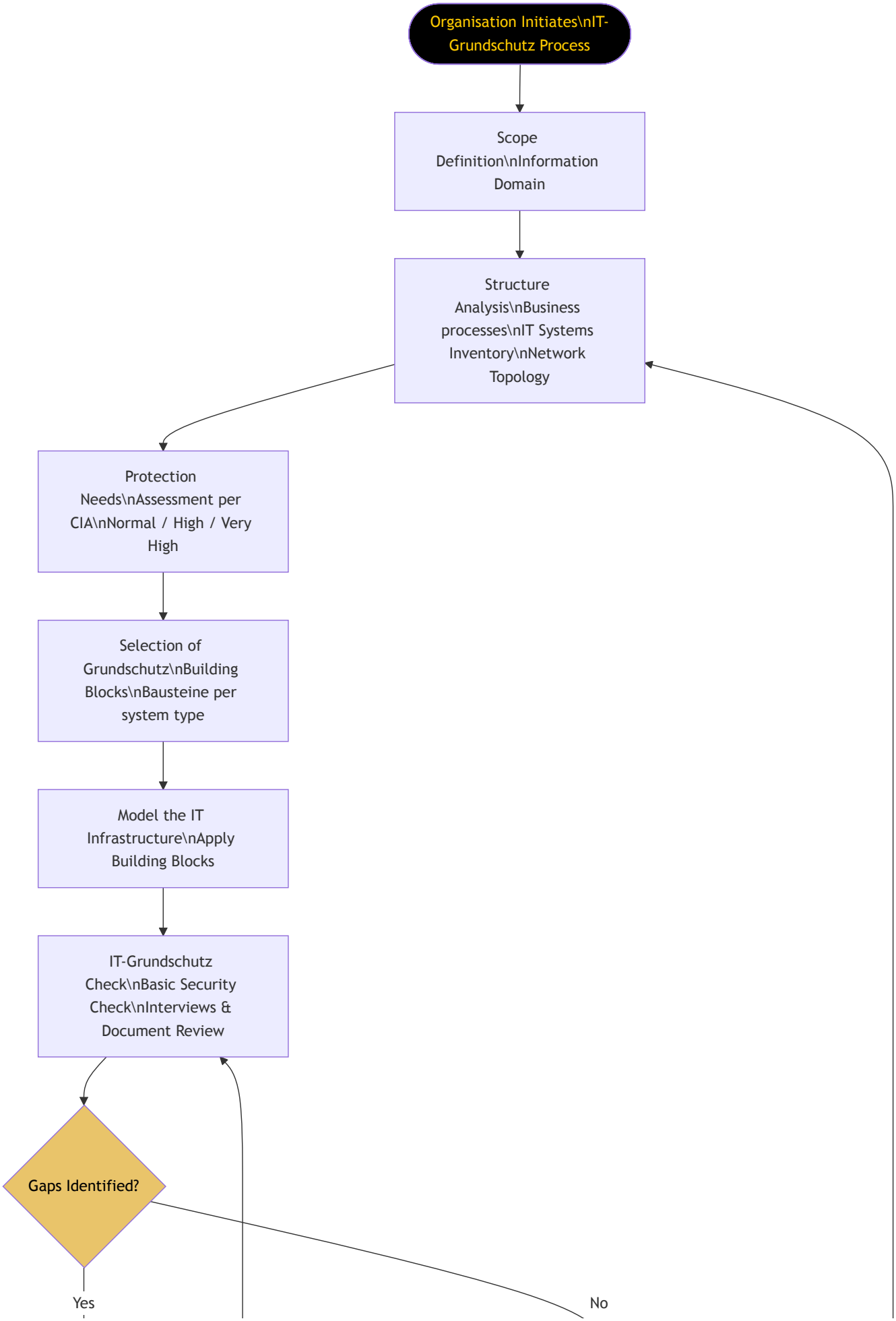


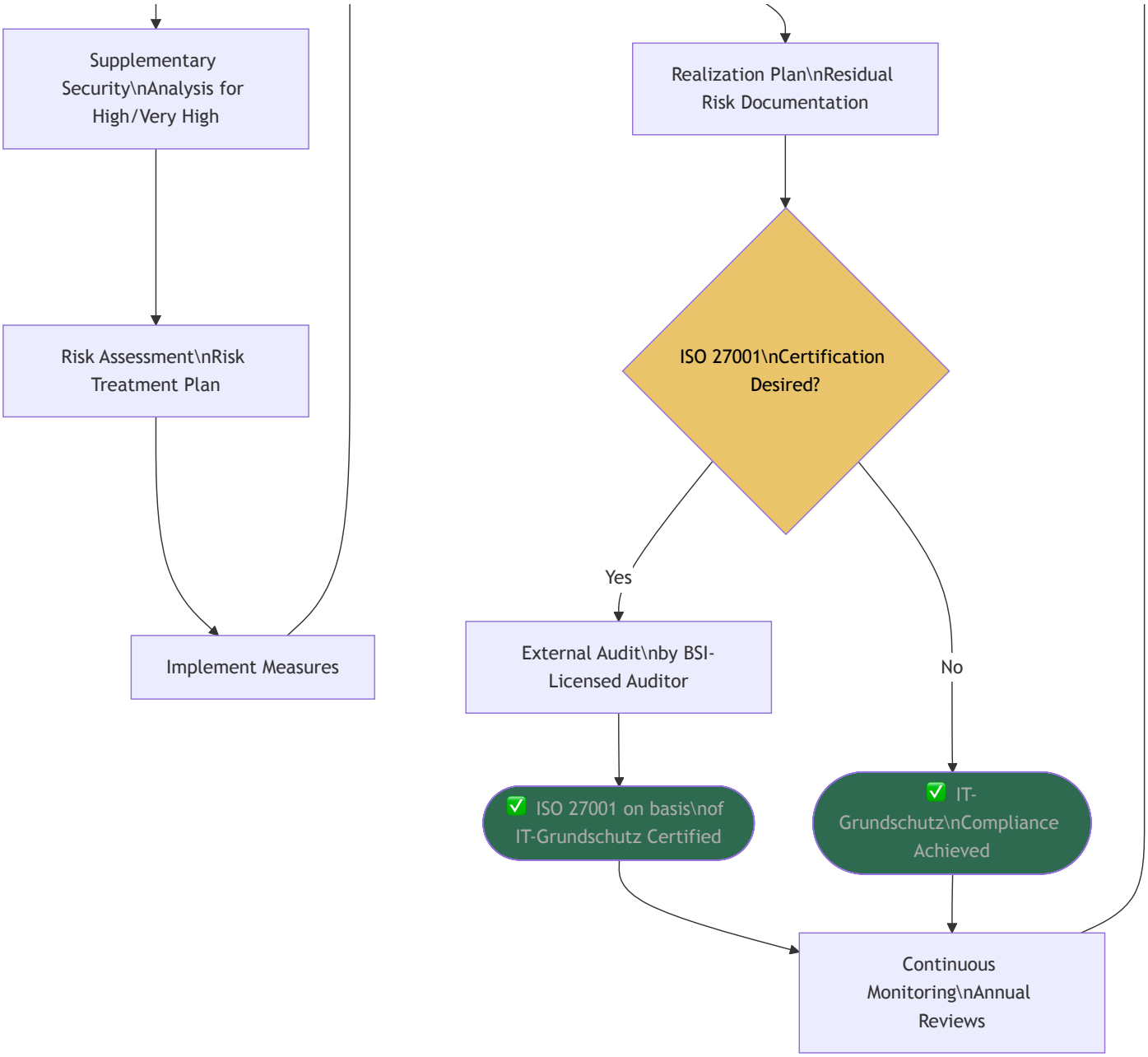
4.3 BSI IT-Grundschutz

Issuing Body: German Federal Office for Information Security (Bundesamt für Sicherheit in der Informationstechnik — BSI) **Applies To:** German federal and state agencies; widely adopted by German private sector and EU entities **Legislative Basis:** German BSI Act (BSIG), updated December 2025 as NIS2 implementation vehicle **Level:** National (Germany) / widely referenced EU-wide

IT-Grundschutz is one of the most detailed IT security frameworks in the world, providing highly prescriptive building blocks (Bausteine) for specific system types and scenarios. Organizations can pursue ISO 27001 certification on the basis of IT-Grundschutz. The framework has three protection requirement levels and four implementation methodologies.

Protection Levels: Normal / High / Very High **Methodologies:** Basic / Standard / Core / Combined


Organisation Initiates\nIT-Grundschutz Process
Scope Definition\nInformation Domain
Structure Analysis\nBusiness processes\nIT Systems Inventory\nNetwork Topology
Protection Needs\nAssessment per CIA\nNormal / High / Very High
Selection of Grundschutz\nBuilding Blocks\nBausteine per system type
Model the IT Infrastructure\nApply Building Blocks
IT-Grundschutz Check\nBasic Security Check\nInterviews & Document Review
Gaps Identified?
Yes
No



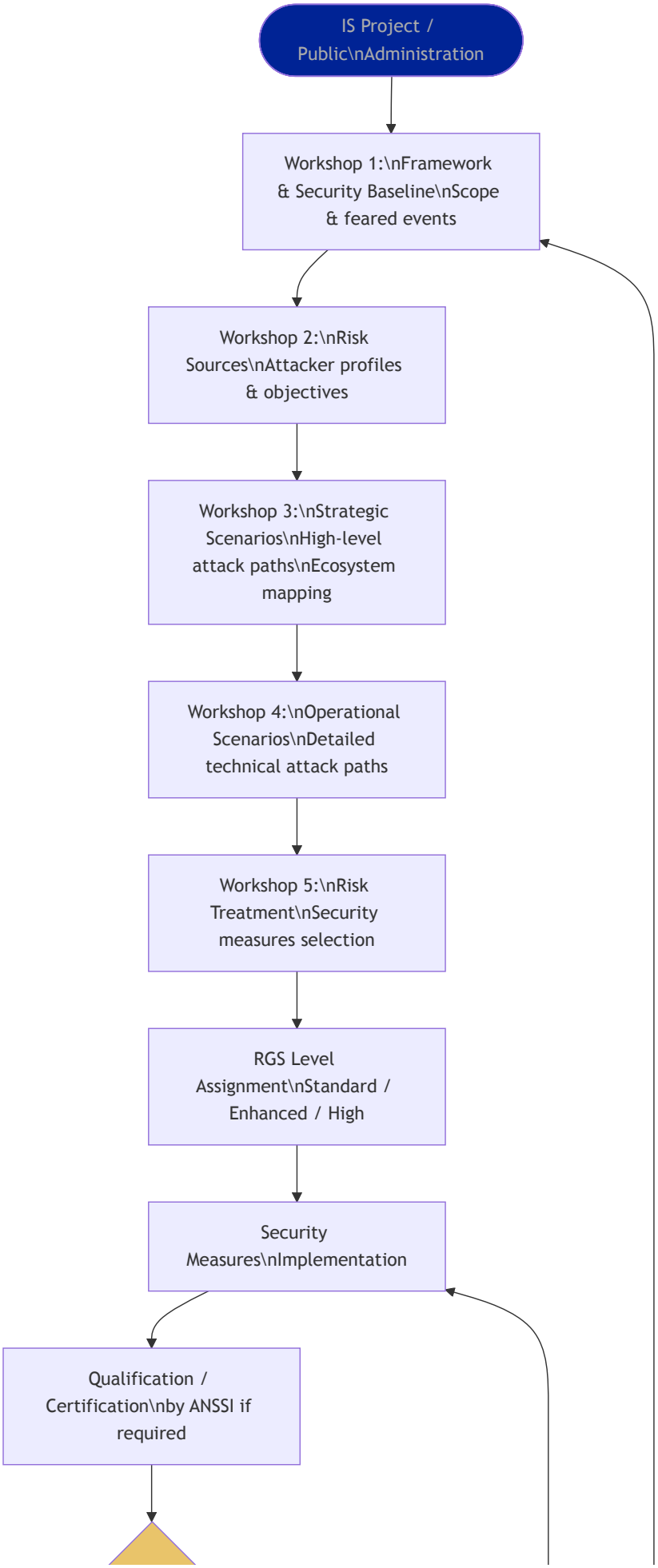
4.4 ANSSI RGS / EBIOS RM (France)

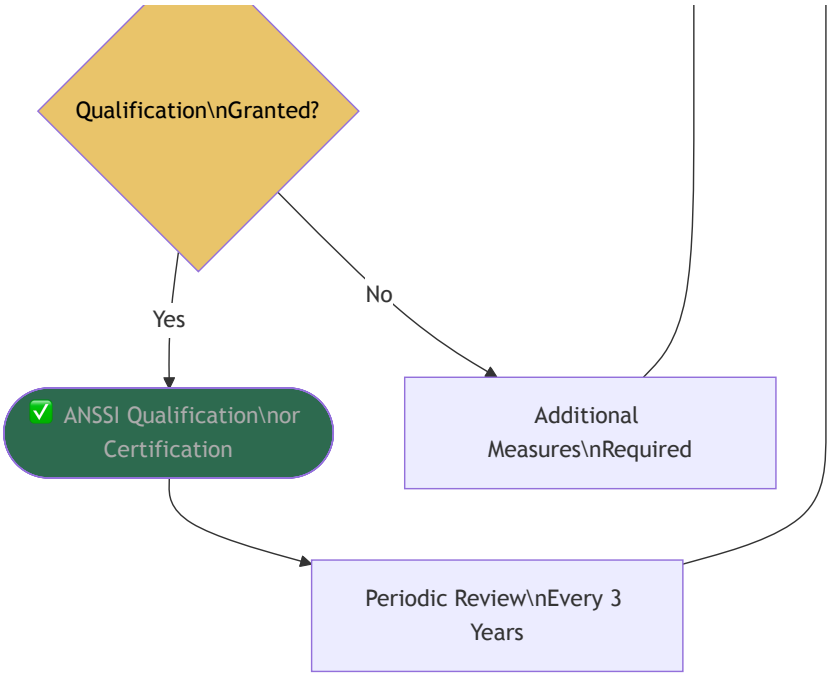
Issuing Body: Agence nationale de la sécurité des systèmes d'information (ANSSI) **Applies To:** French public administrations and operators of vital importance (OIV) **Legislative Basis:** *Ordonnance n° 2005-1516 (RGS)* , *Loi de Programmation Militaire (LPM)* for OIVs **Level:** National (France)

The RGS (Référentiel Général de Sécurité) defines security levels for information systems used by French public services. EBIOS Risk Manager (EBIOS RM) is the companion risk assessment methodology, jointly developed by ANSSI and the Club EBIOS. EBIOS RM is one of the most sophisticated risk assessment methodologies available, using attack path scenarios and ecosystem analysis.

RGS Security Levels: Standard / Enhanced / High **EBIOS RM Workshops:**

1. Framework and Security Baseline
2. Risk Sources and Objectives
3. Strategic Scenarios
4. Operational Scenarios
5. Risk Treatment


IS Project / Public\nAdministration
Workshop 1:\nFramework & Security Baseline\nScope & feared events
Workshop 2:\nRisk Sources\nAttacker profiles & objectives
Workshop 3:\nStrategic Scenarios\nHigh-level attack paths\nEcosystem mapping
Workshop 4:\nOperational Scenarios\nDetailed technical attack paths
Workshop 5:\nRisk Treatment\nSecurity measures selection
RGS Level Assignment\nStandard / Enhanced / High
Security Measures\nImplementation
Qualification / Certification\nby ANSSI if required



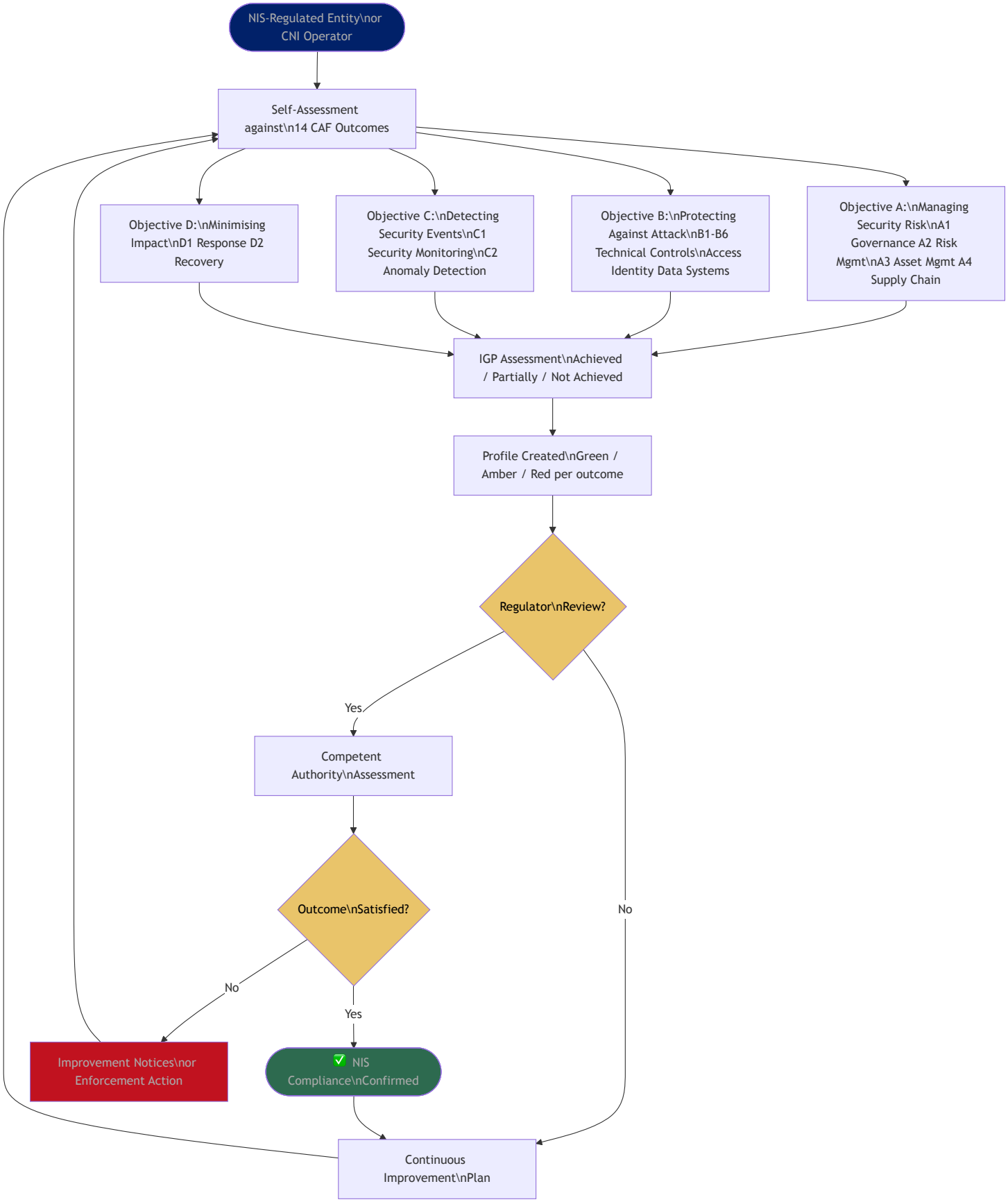
4.5 NCSC CAF — Cyber Assessment Framework (UK)

Issuing Body: UK National Cyber Security Centre (NCSC) **Applies To:** UK Critical National Infrastructure operators, NIS-regulated entities, government departments **Legislative Basis:** UK *NIS Regulations 2018* , *Network and Information Systems (Amendment etc.) (EU Exit) Regulations 2019* **Level:** National (United Kingdom)

The CAF provides a structured set of 14 cybersecurity outcomes across 4 objectives. Regulators and operators use it to assess security posture against defined Indicators of Good Practice (IGPs). The CAF is outcome-focused rather than prescriptive — organizations determine how to achieve each outcome based on their context.

Four Objectives:

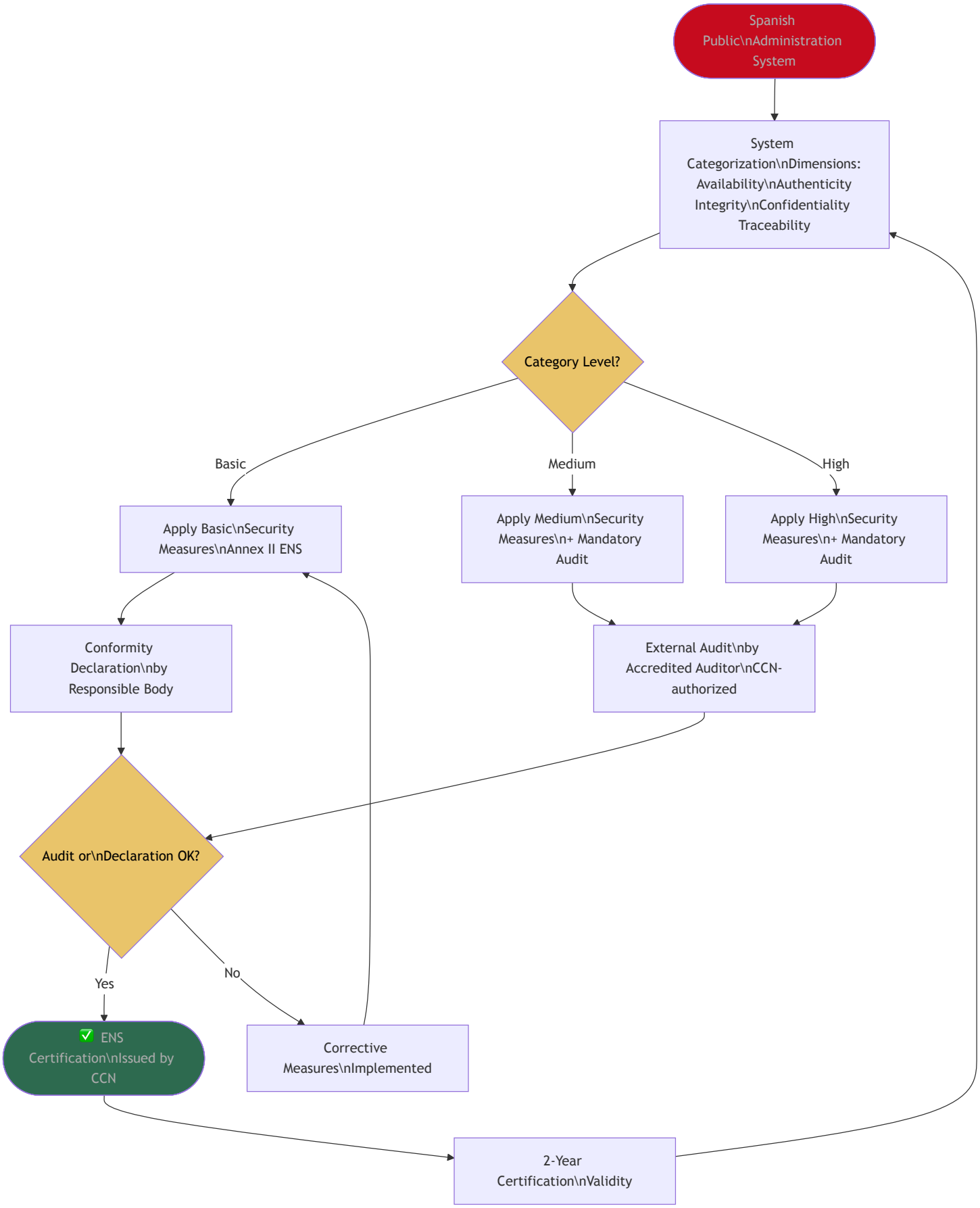
- A: Managing Security Risk
- B: Protecting Against Cyber Attack
- C: Detecting Cyber Security Events
- D: Minimising the Impact of Cyber Security Incidents


NIS-Regulated Entity\nor CNI Operator
Self-Assessment against\n14 CAF Outcomes
Objective D:\nMinimising Impact\nD1 Response D2 Recovery
Objective C:\nDetecting Security Events\nC1 Security Monitoring\nC2 Anomaly Detection
Objective B:\nProtecting Against Attack\nB1-B6 Technical Controls\nAccess Identity Data Systems
Objective A:\nManaging Security Risk\nA1 Governance A2 Risk Mgmt\nA3 Asset Mgmt A4 Supply Chain
IGP Assessment\nAchieved / Partially / Not Achieved
Profile Created\nGreen / Amber / Red per outcome
Regulator\nReview?
Yes
No
Competent Authority\nAssessment
Outcome\nSatisfied?
No
Yes
Improvement Notices\nor Enforcement Action
✅ NIS Compliance\nConfirmed
Continuous Improvement\nPlan

4.6 Spain ENS — Esquema Nacional de Seguridad

Issuing Body: Centro Criptológico Nacional (CCN-CERT) / CPSTIC **Applies To:** All Spanish public administrations and their technology suppliers **Legislative Basis:** Royal Decree 311/2022 (updated ENS), transposing NIS2 obligations for public sector **Level:** National (Spain)

ENS defines security categories (Basic / Medium / High) and prescribes mandatory security measures through the CCN-STIC guide series. It is one of the most prescriptive national frameworks in Europe. Technology providers to the Spanish public sector must obtain ENS certification. Audits are mandatory for Medium and High category systems.


Spanish Public\nAdministration System
System Categorization\nDimensions: Availability\nAuthenticity Integrity\nConfidentiality Traceability
Category Level?
Basic
Medium
High
Apply Basic\nSecurity Measures\nAnnex II ENS
Apply Medium\nSecurity Measures\n+ Mandatory Audit
Apply High\nSecurity Measures\n+ Mandatory Audit
Conformity Declaration\nby Responsible Body
External Audit\nby Accredited Auditor\nCCN-authorized
Audit or\nDeclaration OK?
Yes
No
ENS Certification\nIssued by CCN
Corrective Measures\nImplemented
2-Year Certification\nValidity

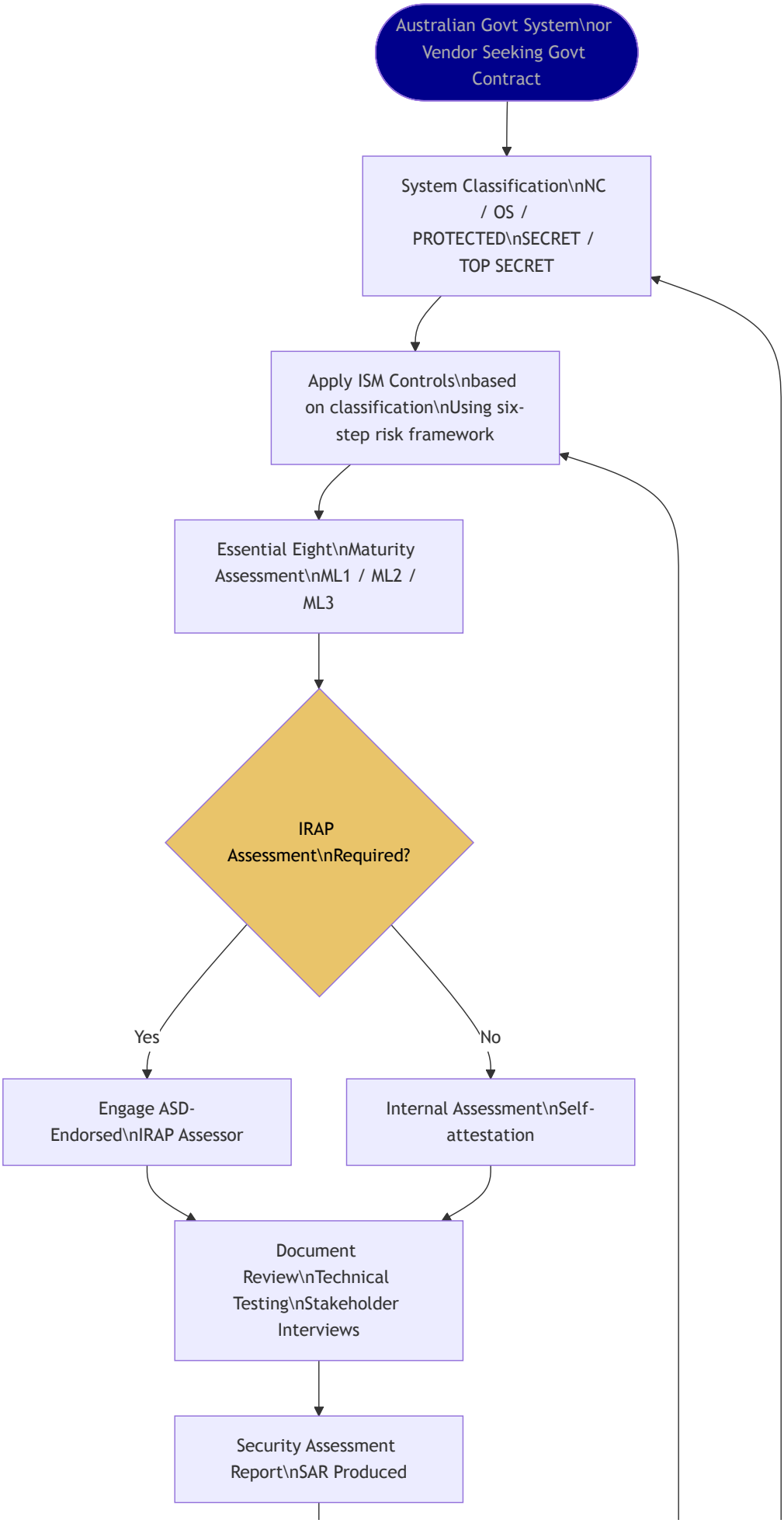
5. Australia

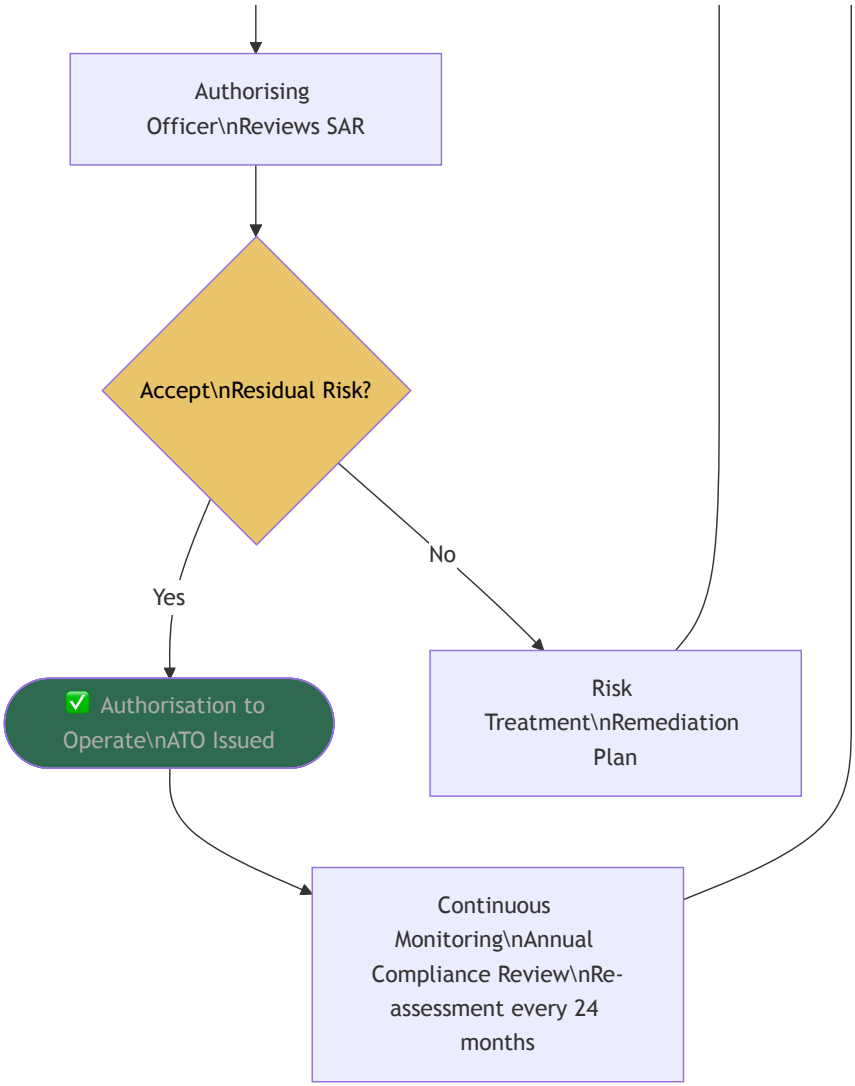
5.1 ASD Information Security Manual (ISM) + IRAP

Issuing Body: Australian Signals Directorate (ASD) **Applies To:** All Australian Commonwealth agencies, state governments, and private sector entities handling government data **Legislative Basis:** *Intelligence Services Act 2001* , *Security of Critical Infrastructure Act 2018 (SOCI Act)* **Level:** Federal + State (widely)

The ISM is Australia's primary IT security framework — the direct equivalent of ITSG-33. It is updated multiple times per year. Organizations use the ISM's six-step risk management approach and apply controls based on their system's classification level. IRAP (Infosec Registered Assessors Program) is the formal assessment program — ASD-endorsed assessors independently evaluate systems against ISM controls and produce a Security Assessment Report (SAR), which feeds into an Authorisation to Operate (ATO) issued by the system's Authorising Officer.

Classification Levels: Non-Classified / OFFICIAL: Sensitive / PROTECTED / SECRET / TOP SECRET **IRAP Assessment Types:** PROTECTED-level (most common commercial), SECRET/TS (government-led)


Australian Govt System\nor Vendor Seeking Govt Contract
System Classification\nNC / OS / PROTECTED\nSECRET / TOP SECRET
Apply ISM Controls\nbased on classification\nUsing six-step risk framework
Essential Eight\nMaturity Assessment\nML1 / ML2 / ML3
IRAP Assessment\nRequired?
Yes
No
Engage ASD-Endorsed\nIRAP Assessor
Internal Assessment\nSelf-attestation
Document Review\nTechnical Testing\nStakeholder Interviews
Security Assessment Report\nSAR Produced



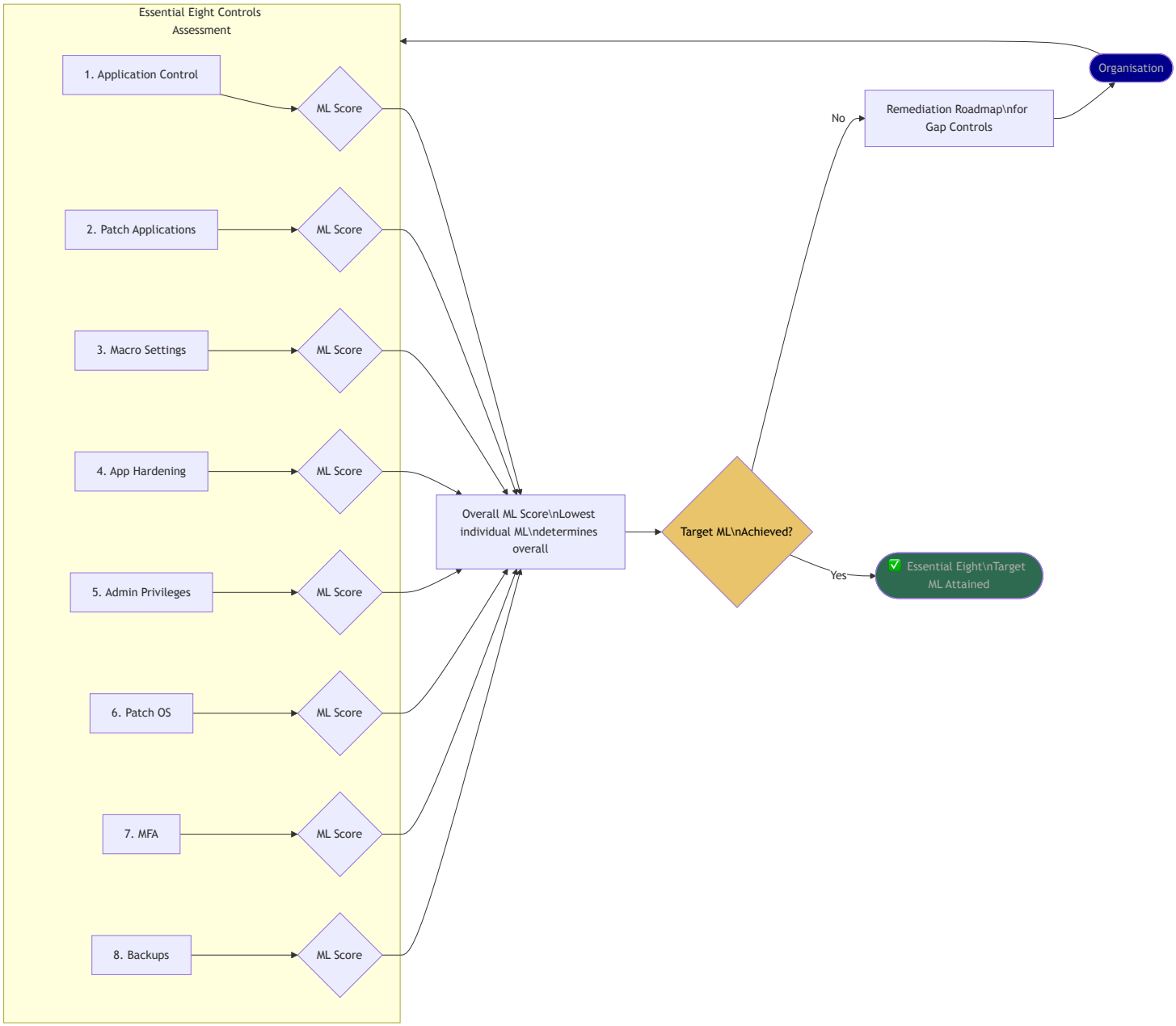
5.2 ASD Essential Eight

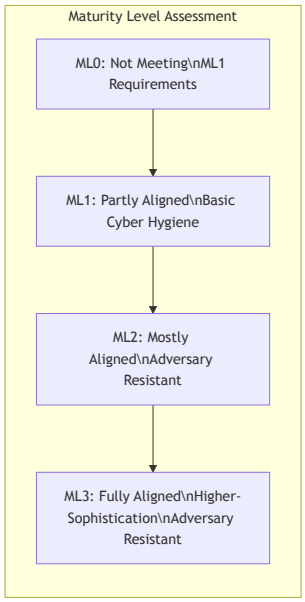
Issuing Body: Australian Signals Directorate (ASD) **Applies To:** All Australian organisations with internet-connected IT networks; mandatory baseline for government **Legislative Basis:** *SOCI Act 2018* for critical infrastructure; mandatory for Commonwealth via PSPF **Level:** Federal + widely adopted by state/territory governments

The Essential Eight is a prioritized subset of ISM controls representing the most effective baseline mitigations. It uses a three-level maturity model (ML1/ML2/ML3). Unlike the full ISM, the Essential Eight is often used as a standalone assessment target, particularly by state governments and private sector entities not requiring full IRAP.

The Eight Strategies:

1. Application Control
2. Patch Applications
3. Configure Microsoft Office Macro Settings
4. User Application Hardening
5. Restrict Administrative Privileges
6. Patch Operating Systems
7. Multi-Factor Authentication
8. Regular Backups


Essential Eight Controls Assessment
1. Application Control
2. Patch Applications
3. Macro Settings
4. App Hardening
5. Admin Privileges
6. Patch OS
7. MFA
8. Backups
ML Score
ML Score
ML Score
ML Score
ML Score
ML Score
ML Score
ML Score
Overall ML Score\nLowest individual ML\ndetermines overall
Target ML\nAchieved?
No
Remediation Roadmap\nfor Gap Controls
Organisation
Yes
Essential Eight\nTarget ML Attained


Maturity Level Assessment
ML0: Not Meeting\nML1 Requirements
ML1: Partly Aligned\nBasic Cyber Hygiene
ML2: Mostly Aligned\nAdversary Resistant
ML3: Fully Aligned\nHigher-Sophistication\nAdversary Resistant

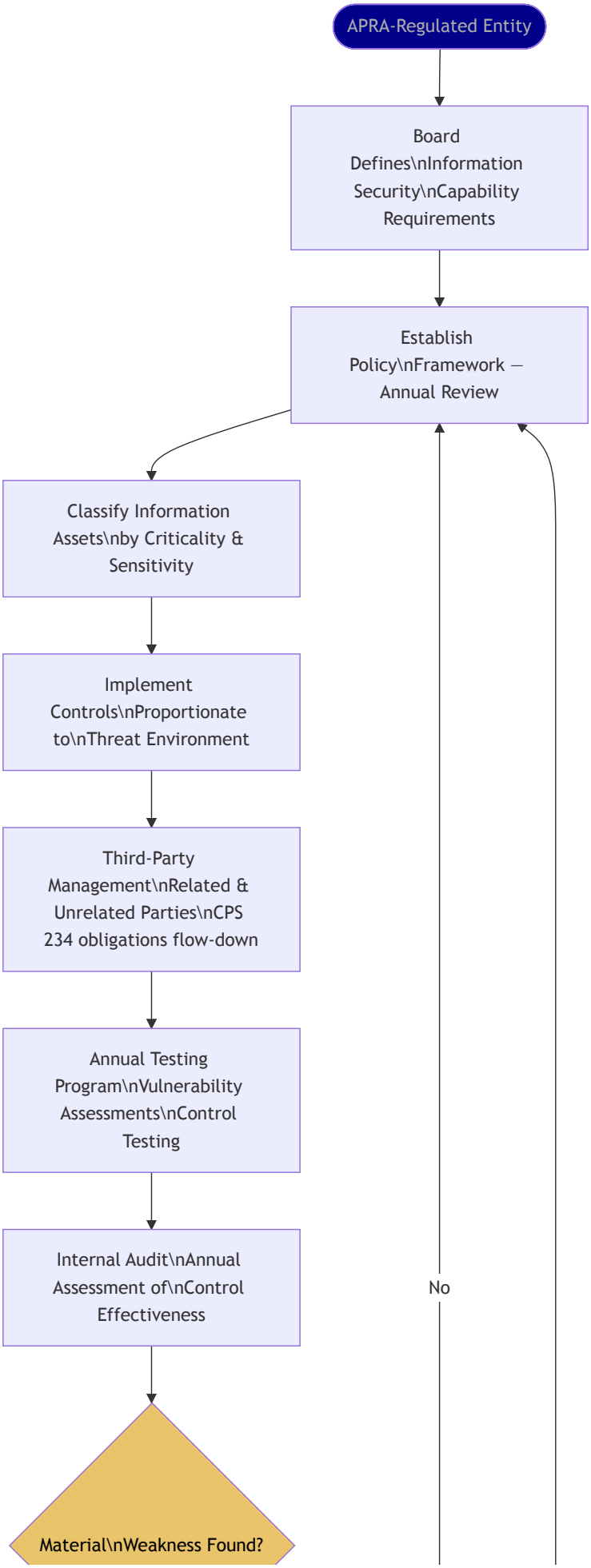
5.3 APRA CPS 234 — Information Security

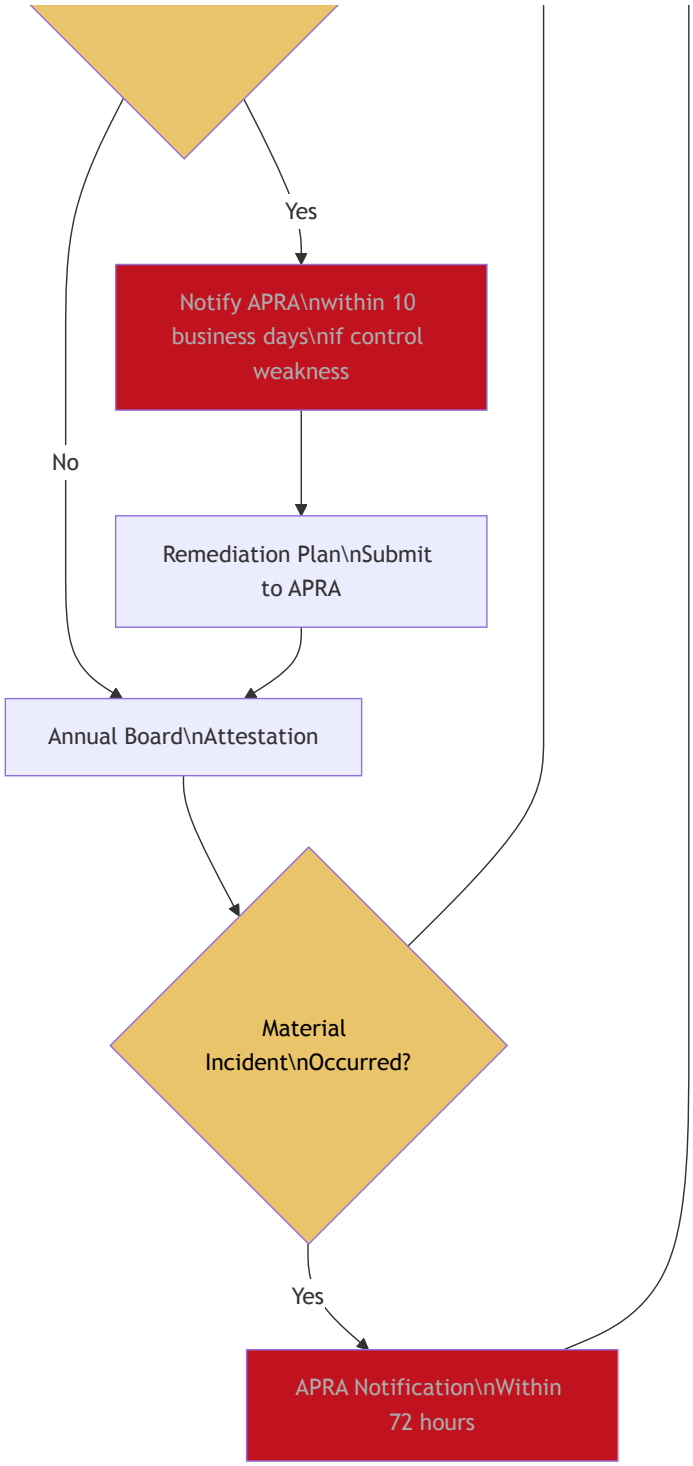
Issuing Body: Australian Prudential Regulation Authority (APRA) **Applies To:** All APRA-regulated entities — banks, insurers, superannuation funds, non-operating holding companies **Legislative Basis:** *Banking Act 1959* , *Insurance Act 1973* , *Superannuation Industry (Supervision) Act 1993* **Level:** National / Financial Sector

CPS 234 is one of the most prescriptive financial sector IT security standards globally. It requires boards to be accountable for information security, mandates a three-lines-of-defence model, requires annual internal audits of security controls, and mandates notification to APRA of material security incidents within 72 hours. Companion guidance CPG 234 provides implementation detail.

Key Requirements:

- Information security capability proportionate to vulnerabilities and threats
- Policy framework maintained and reviewed annually
- Controls testing at least annually or after significant changes
- Third-party (related and unrelated) information security obligations
- Internal audit assessment of control effectiveness


APRA-Regulated Entity
Board Defines\nInformation Security\nCapability Requirements
Establish Policy\nFramework — Annual Review
Classify Information Assets\nby Criticality & Sensitivity
Implement Controls\nProportionate to\nThreat Environment
Third-Party Management\nRelated & Unrelated Parties\nCPS 234 obligations flow-down
Annual Testing Program\nVulnerability Assessments\nControl Testing
Internal Audit\nAnnual Assessment of\nControl Effectiveness
Material\nWeakness Found?
No

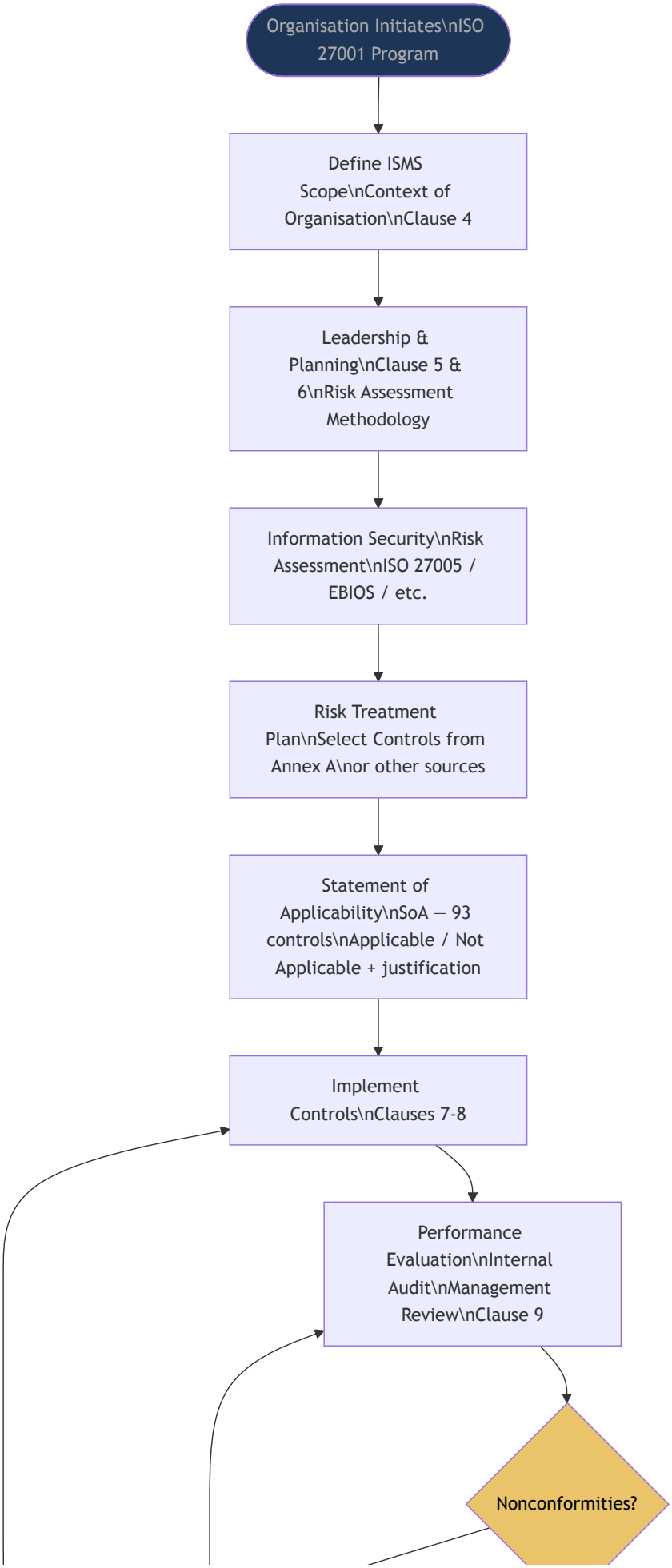


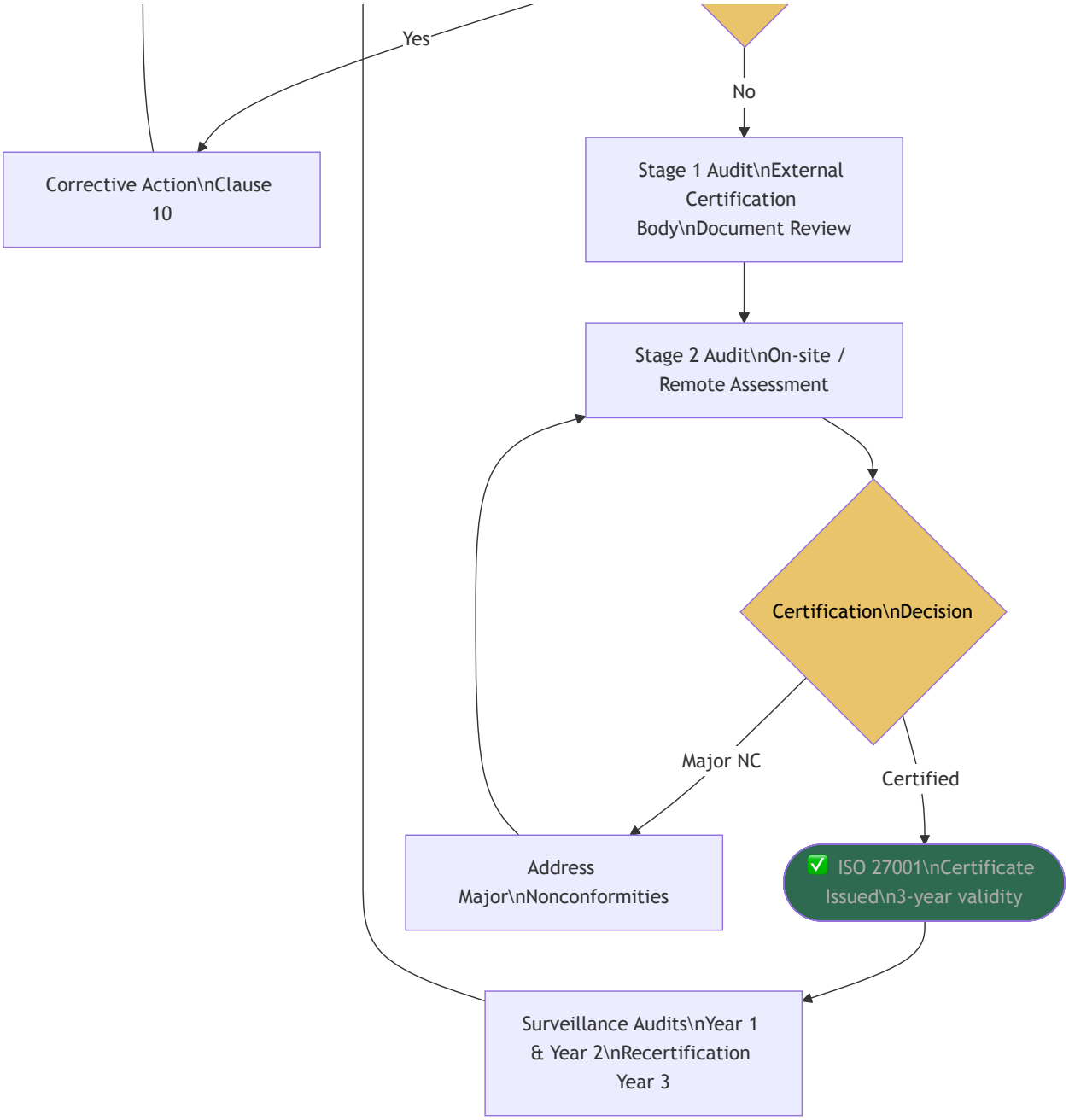
6. Cross-Sector / International

6.1 ISO/IEC 27001:2022 — ISMS

Issuing Body: International Organization for Standardization (ISO) / IEC **Applies To:** Any organisation globally; referenced in legislation across all four regions **Legislative Standing:** Voluntary standard; referenced as compliance mechanism in NIS2, ENS, BSI Act, OSFI B-13, APRA CPS 234, and many procurement regulations **Level:** International

ISO 27001 defines requirements for an Information Security Management System (ISMS). Annex A contains 93 controls across 4 themes (Organizational, People, Physical, Technological). Certification is achieved through accredited external audit. ISO 27002 provides implementation guidance. ISO 27005 addresses information security risk management methodology.

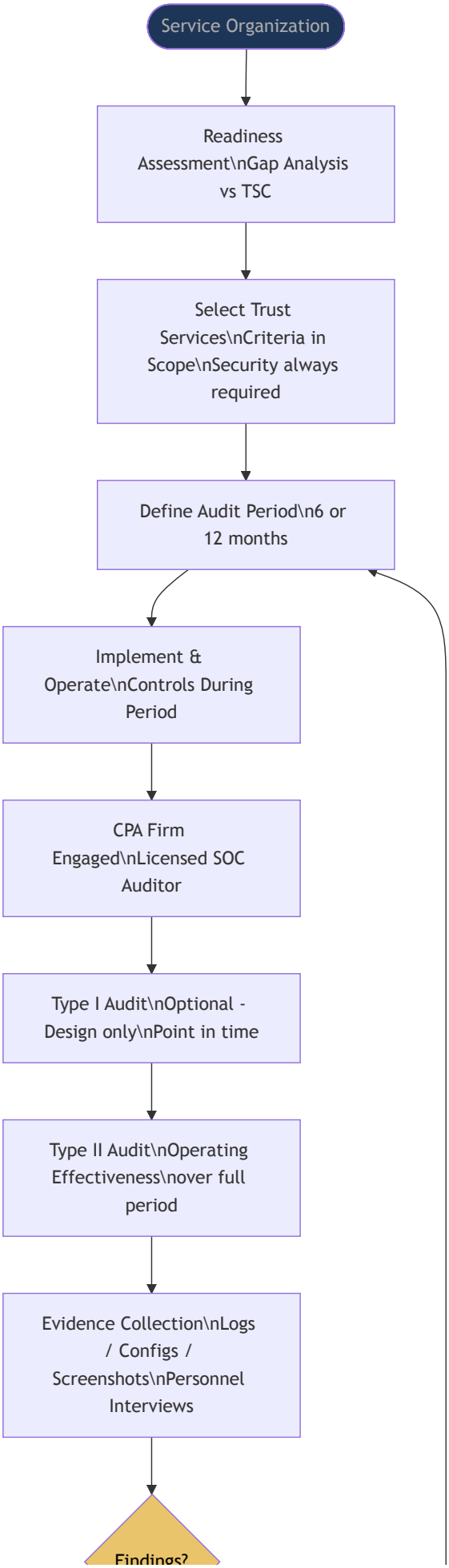

Organisation Initiates\nISO 27001 Program
Define ISMS Scope\nContext of Organisation\nClause 4
Leadership & Planning\nClause 5 & 6\nRisk Assessment Methodology
Information Security\nRisk Assessment\nISO 27005 / EBIOS / etc.
Risk Treatment Plan\nSelect Controls from Annex A\nor other sources
Statement of Applicability\nSoA — 93 controls\nApplicable / Not Applicable + justification
Implement Controls\nClauses 7-8
Performance Evaluation\nInternal Audit\nManagement Review\nClause 9
Nonconformities?

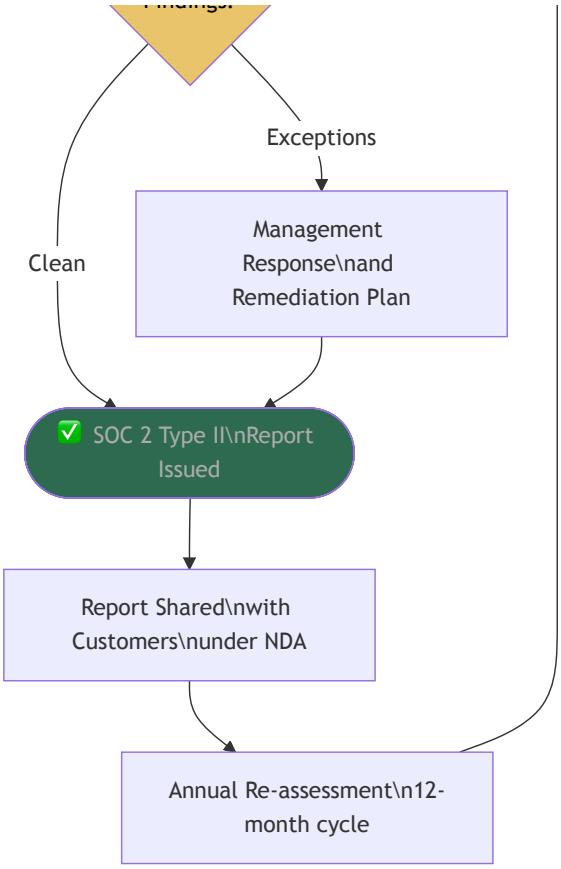


6.2 SOC 2 Type II

Issuing Body: American Institute of CPAs (AICPA) **Applies To:** Technology and cloud service providers; common requirement in US, Canada, Australia **Legislative Standing:** Not legally mandated but contractually required by major enterprises and governments; referenced in ITSP.50.105 (CCCS) **Level:** International / Service Organizations

SOC 2 evaluates service organization controls against five Trust Services Criteria: Security, Availability, Processing Integrity, Confidentiality, and Privacy. Type II covers a period of 6-12 months, making it an evidence of operational effectiveness over time. Commonly required by GC departments, US federal contractors, and APRA-regulated entities assessing their cloud vendors.

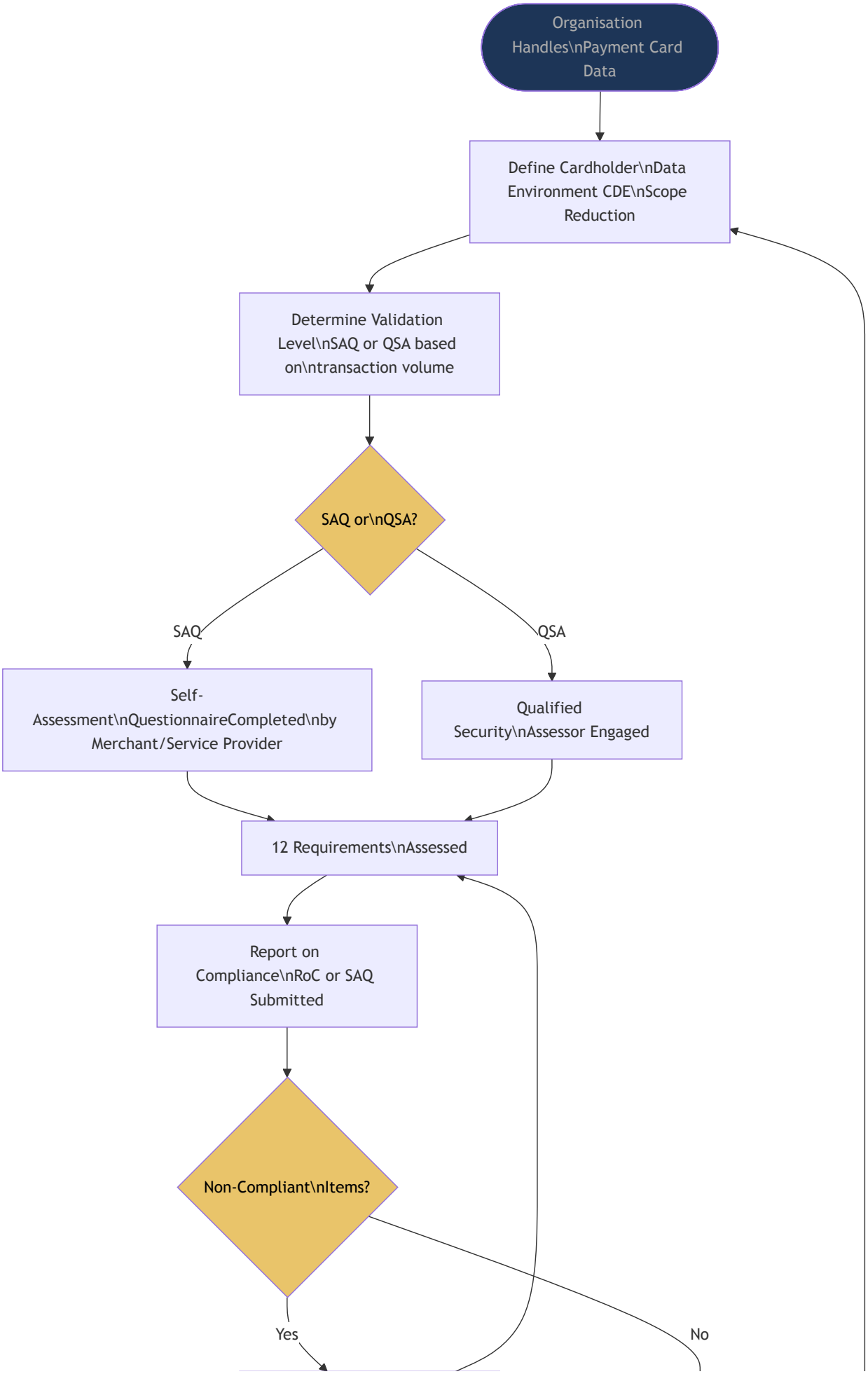

Service Organization
Readiness Assessment\nGap Analysis vs TSC
Select Trust Services\nCriteria in Scope\nSecurity always required
Define Audit Period\n6 or 12 months
Implement & Operate\nControls During Period
CPA Firm Engaged\nLicensed SOC Auditor
Type I Audit\nOptional - Design only\nPoint in time
Type II Audit\nOperating Effectiveness\nover full period
Evidence Collection\nLogs / Configs / Screenshots\nPersonnel Interviews
Findings?

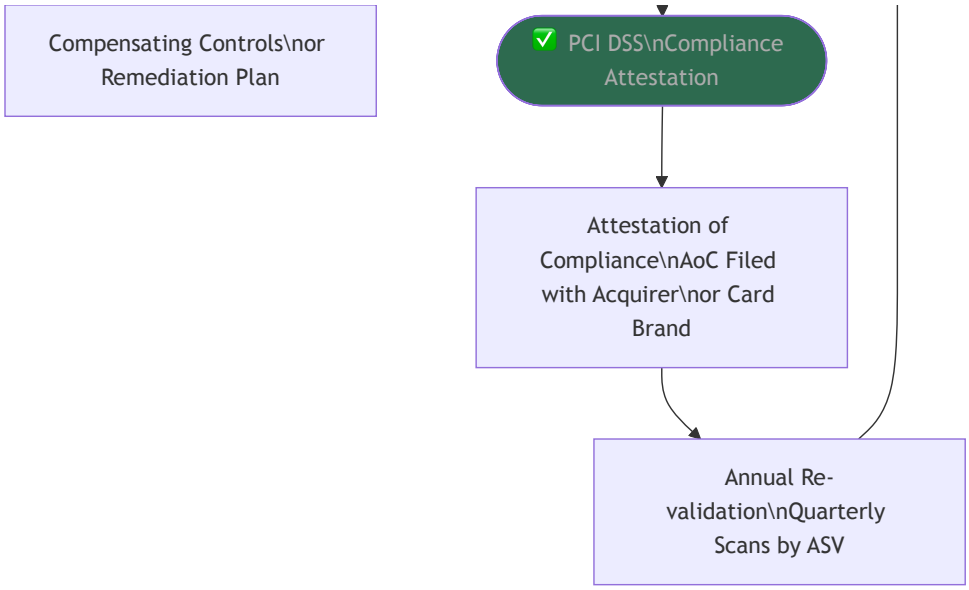


6.3 PCI DSS 4.0

Issuing Body: PCI Security Standards Council (Visa, Mastercard, Amex, Discover, JCB) **Applies To:** Any organisation storing, processing, or transmitting payment card data globally **Legislative Standing:** Contractual requirement enforced by card brands; legislated in some US states (Nevada, Minnesota, Washington); referenced in Canadian, Australian, and EU payment regulations **Level:** Global / Payment Sector

PCI DSS v4.0 (March 2022) defines 12 requirements across 6 control objectives for protecting cardholder data. Organizations validate compliance either via Self-Assessment Questionnaire (SAQ) or Qualified Security Assessor (QSA) audit depending on transaction volume.

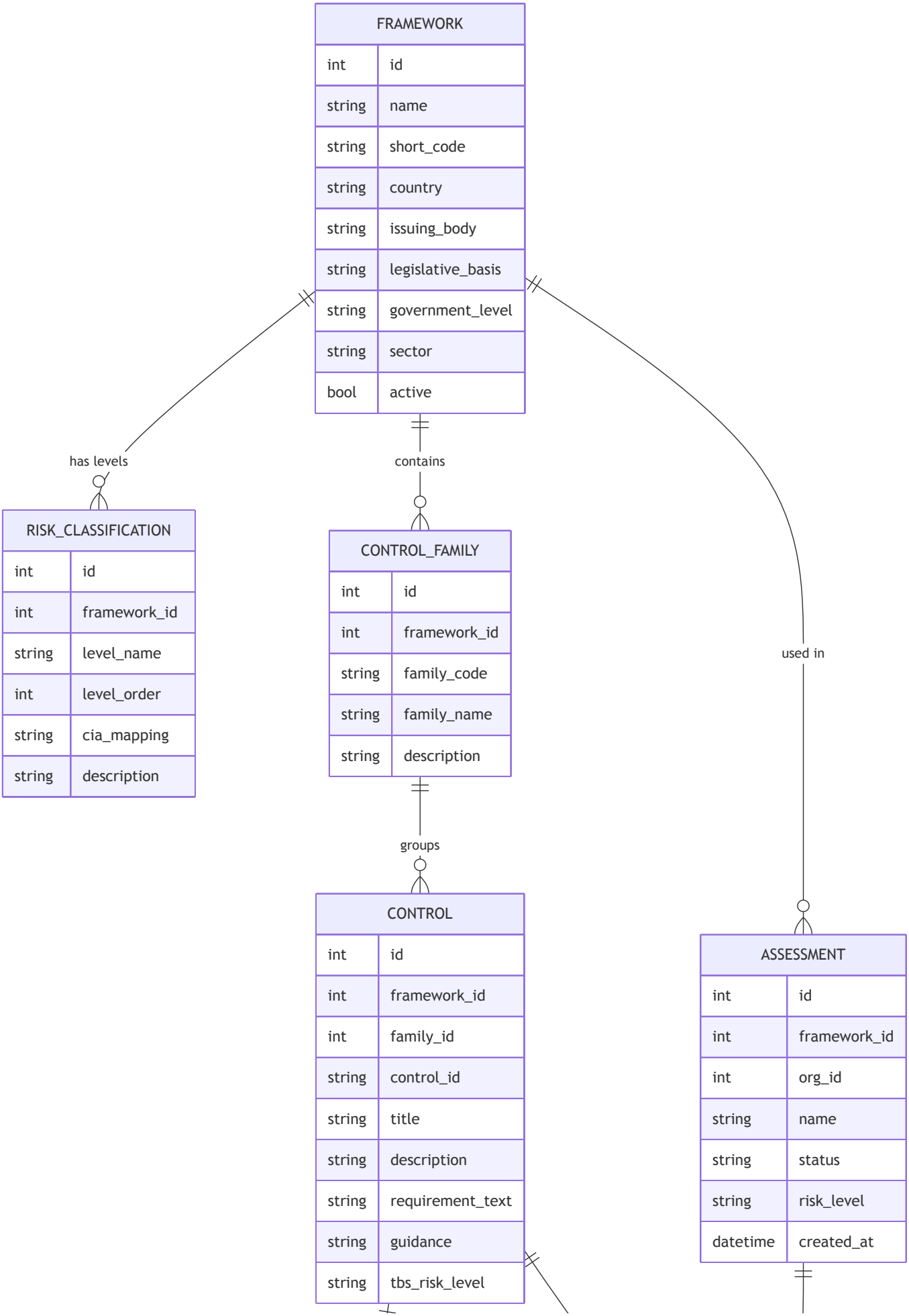

Organisation Handles\nPayment Card Data
Define Cardholder\nData Environment CDE\nScope Reduction
Determine Validation Level\nSAQ or QSA based on\ntransaction volume
SAQ or\nQSA?
SAQ
QSA
Self-Assessment\nQuestionnaireCompleted\nby Merchant/Service Provider
Qualified Security\nAssessor Engaged
12 Requirements\nAssessed
Report on Compliance\nRoC or SAQ Submitted
Non-Compliant\nItems?
Yes
No

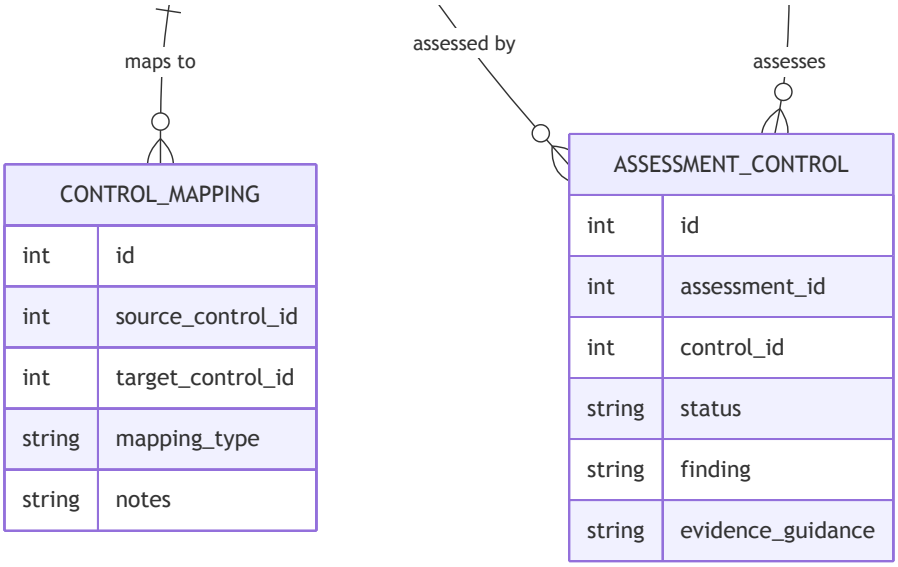


7. Multi-Framework Data Model

How to Support Multiple Frameworks in One Tool

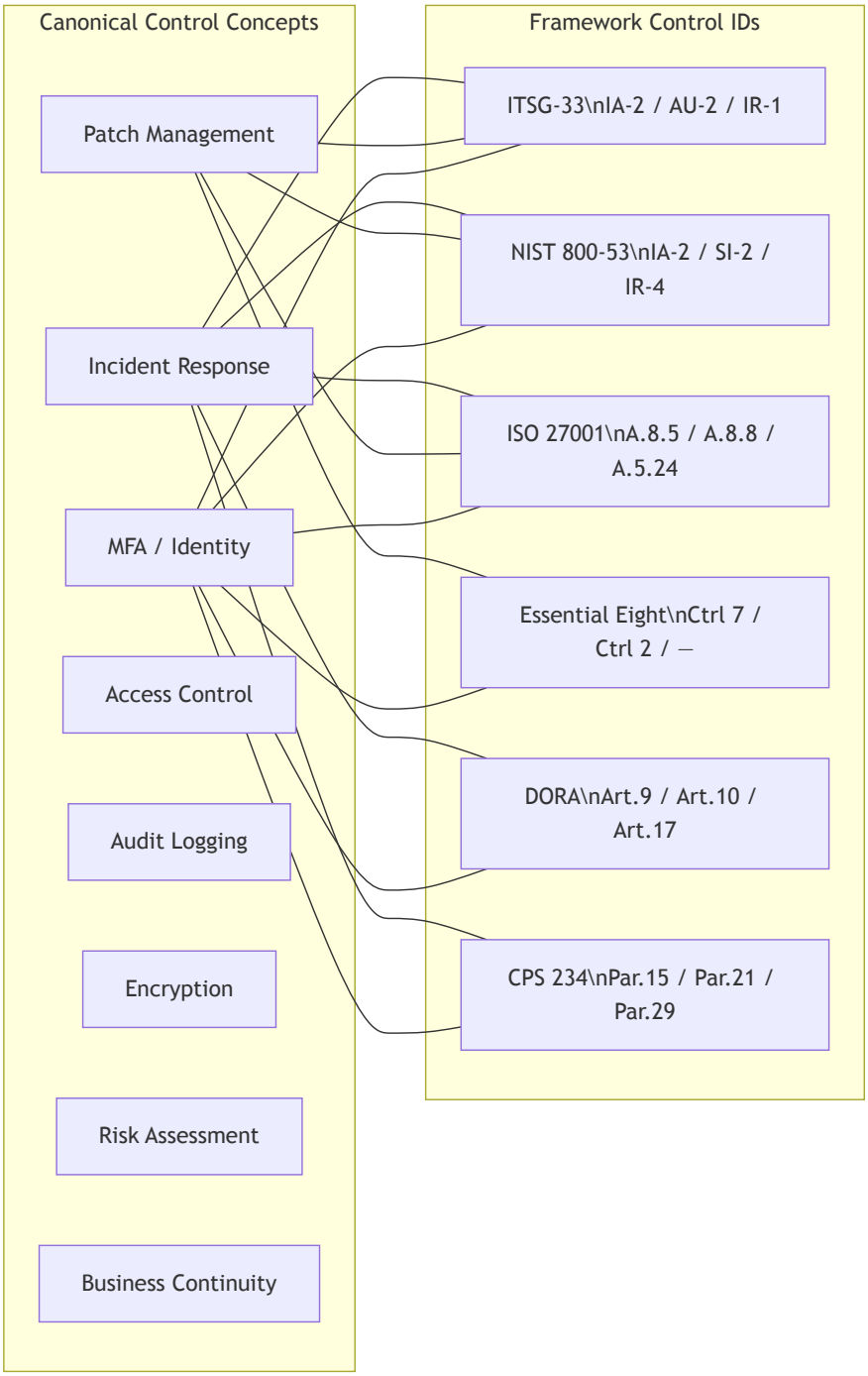
The key insight is that all frameworks share the same SA&A lifecycle with three variables: **control catalogue** , **risk classification scheme** , and **authorization terminology** . Your existing schema only needs extension, not replacement.


FRAMEWORK
int id
string name
string short_code
string country
string issuing_body
string legislative_basis
string government_level
string sector
bool active
has levels
contains
used in
RISK_CLASSIFICATION
int id
int framework_id
string level_name
int level_order
string cia_mapping
string description
CONTROL_FAMILY
int id
int framework_id
string family_code
string family_name
string description
groups
CONTROL
int id
int framework_id
int family_id
string control_id
string title
string description
string requirement_text
string guidance
string tbs_risk_level
ASSESSMENT
int id
int framework_id
int org_id
string name
string status
string risk_level
datetime created_at



Framework-to-Framework Control Mapping

Since many frameworks share control concepts (e.g., MFA appears in ITSG-33 IA-2, NIST IA-2, ISO 27001 A.8.5, Essential Eight Control 7, CPS 234, DORA Article 9), the `CONTROL_MAPPING` table allows cross-framework reuse:



Authorization Terminology Mapping

Your Tool Term	ITSG-33	NIST RMF	IRAP/ISM	NIS2/DORA	ISO 27001	ENS
ATO	ATO Letter	Authorization to Operate	Authorisation to Operate	—	Certificate	ENS Certificate
iATO	iATO	IATT	Interim ATO	—	—	—
Authorizing Official	AO	AO	Authorising Officer	Board / NCA	Top Management	Responsible Body
Security Assessment	SAR	SAR	SAR	Audit Report	Internal Audit Report	Audit Report

Your Tool Term	ITSG-33	NIST RMF	IRAP/ISM	NIS2/DORA	ISO 27001	ENS
Report						
POA&M	POA&M	POA&M	Risk Treatment Plan	Corrective Action Plan	Corrective Action	Improvement Plan
Control Profile	Control Profile	Baseline	SCM	National Framework	Annex A / SoA	ENS Annex II
Risk Level	Low/Med/High (CIA)	Low/Mod/High (FIPS 199)	NC/OS/P/S/TS	Essential/Important	Accepted/Treated	Basic/Medium/High

8. Framework Comparison Matrix

Framework	Country	Level	Sector	Control Count	Risk Levels	ATO Equivalent	Mandatory
ITSG-33	🇨🇦 Canada	Federal	All Govt	~300	Low/Med/High (CIA)	ATO Letter	Yes (GoC)
CCCS Medium (PBMM)	🇨🇦 Canada	Federal	Cloud	~200	PBMM fixed	ATO Letter	Yes (GC Cloud)
OSFI B-13	🇨🇦 Canada	Federal	Finance	Outcomes-based	—	Supervisory Review	Yes (FRFIs)
NIST RMF / 800-53	🇺🇸 USA	Federal	All	~1000+	Low/Mod/High	ATO	Yes (FISMA)
FedRAMP	🇺🇸 USA	Federal	Cloud	~325 Mod	Low/Mod/High	P-ATO / ATO	Yes (Fed Cloud)
CMMC 2.0	🇺🇸 USA	Federal	Defense	17–110	L1/L2/L3	Certification	Yes (DoD contracts)
StateRAMP	🇺🇸 USA	State	Cloud	~325	Low/Mod/High	Provisional Auth.	Yes (member states)
NYDFS 500	🇺🇸 USA	State	Finance	~23 sections	—	Annual Certification	Yes (NY-regulated)
NIS2	🇪🇺 EU	National	18 sectors	Outcome-based	Essential/Important	NCA Compliance	Yes (EU)
DORA	🇪🇺 EU	National	Finance	5 pillars	Significant/Other	ESA Compliance	Yes (EU Finance)
BSI IT-Grundschutz	🇩🇪 Germany	National	All	100+ Bausteine	Normal/High/VH	Certification	Yes (govt)
ANSSI RGS / EBIOS RM	🇫🇷 France	National	Public Admin	3 levels	Standard/Enhanced/High	ANSSI Qualification	Yes (French govt)
NCSC CAF	🇬🇧 UK	National	CNI	14 outcomes	IGP scores	NIS Compliance	Yes (NIS-regulated)

Framework	Country	Level	Sector	Control Count	Risk Levels	ATO Equivalent	Mandatory
ENS (CCN-STIC)	🇪🇸 Spain	National	Public Admin	~75 measures	Basic/Medium/High	ENS Certificate	Yes (Spanish govt)
ASD ISM + IRAP	🇦🇺 Australia	Federal	All Govt	~900+	NC/OS/P/S/TS	ATO	Yes (Commonwealth)
ASD Essential Eight	🇦🇺 Australia	Federal+	All	8 strategies	ML1/ML2/ML3	ML Attainment	Yes (PSPF)
APRA CPS 234	🇦🇺 Australia	National	Finance	Outcomes-based	Criticality tiers	Board Attestation	Yes (APRA entities)
ISO/IEC 27001:2022	🌐 Global	—	All	93 (Annex A)	Risk-based	Certificate	Voluntary/Referenced
SOC 2 Type II	🌐 Global	—	Service Orgs	5 TSC	—	SOC 2 Report	Contractual
PCI DSS 4.0	🌐 Global	—	Payments	12 requirements	—	AoC	Contractual

Document version 1.0 — prepared for SA&A tool multi-framework expansion planning Frameworks included: IT security assessment lifecycles with control catalogues, risk categorization, evidence collection, and authorization/certification outputs only.